

Bistability of the climate around the habitable zone: A thermodynamic investigation



Robert Boschi^{a,*}, Valerio Lucarini^{a,b}, Salvatore Pascale^a

^a KlimaCampus, Meteorologisches Institut, Universität Hamburg, Hamburg, Germany

^b Department of Mathematics and Statistics, University of Reading, Reading, UK

ARTICLE INFO

Article history:

Available online 9 April 2013

Keywords:

Atmospheres, dynamics
Extra-solar planets
Rotational dynamics
Terrestrial planets
Solar radiation

ABSTRACT

The goal of this paper is to explore the potential multistability of the climate for a planet around the habitable zone. We apply our methodology to the Earth system, but our investigation has more general relevance. A thorough investigation of the thermodynamics of the climate system is performed for very diverse conditions of energy input and infrared atmosphere opacity. Using PlaSim, an Earth-like general circulation model, the solar constant S^* is modulated between 1160 and 1510 W m^{-2} and the CO_2 concentration, $[\text{CO}_2]$, between 90 and 2880 ppm. It is observed that in such a parameter range the climate is bistable, i.e. there are two coexisting attractors, one characterised by warm, moist climates (W) and one by completely frozen sea surface (Snowball Earth, SB). The tipping points of both the transitions ($W \rightarrow SB$ and $SB \rightarrow W$) are located along straight lines in the $(S^*, \log[\text{CO}_2])$ space. The dynamical and thermodynamical properties – energy fluxes, Lorenz energy cycle, Carnot efficiency, material entropy production – of the W and SB states are very different: W states are dominated by the hydrological cycle and latent heat is prominent in the material entropy production; the SB states are eminently dry climates where heat transport is realised through sensible heat fluxes and entropy mostly generated by dissipation of kinetic energy. We also show that the Carnot efficiency regularly increases towards each transition between W and SB, with a large discontinuous decrease at the point of each transition. Finally, we propose well-defined empirical functions allowing for expressing the global non-equilibrium thermodynamical properties of the system in terms of either the mean surface temperature or the mean planetary emission temperature. While the specific results presented in this paper depend on some characteristics of the Earth system (e.g. rotation rate, position of the continents), this paves the way for the possibility of proposing efficient parameterisations of complex non-equilibrium properties and of practically deducing fundamental properties of a planetary system from a relatively simple observable. As a preliminary result, we obtain that when reducing the rotation rate of the planet by a factor of two, the multistability properties, the quantitative estimators of the thermodynamics of the system, and the approximate parameterisations in terms of the surface of emission temperature are only weakly affected.

© 2013 Published by Elsevier Inc.

1. Introduction

1.1. Planetary atmospheres and extrasolar planets

A very active field of research in astrophysical sciences at the moment, is the observational, model-assisted, and theoretical investigation of extra-solar planetary objects. In less than two decades, the development of new instruments have lead to the detection of the first extra-solar planet in the mid 1990s to more refined observations of the characteristic properties of several hundred planetary bodies. The properties of planets vary greatly, in terms of their compositions – gaseous or rocky – nature of their atmosphere, and of their size-ranging in several orders of magnitude. Astronomical and astrophysical factors of great relevance include

* Corresponding author.

E-mail address: robert.boschi@zmaw.de (R. Boschi).

the temperature and the intensity of the radiation emitted by the parent star, the orbital parameters, and whether or not the planet is tidally locked. A great deal of effort has been directed at constraining the combinations of physical configurations potentially compatible with life (habitable zone), that is to say with the possibility of observing water, either prominently or at least partially in liquid form at the surface. Obviously, the so-called habitable zone is the setting where we hope to find forms of extraterrestrial life, or at least life analogous to that found on Earth. Recently, ESOs HARPS planet finder estimated that just in the Milky Way billions of habitable Earth-like rocky planets could orbit around faint red dwarfs, with in the order of one hundred being in the immediate vicinity of our Solar System (Bonfils et al., 2013). We refer the reader to the website (<http://exoplanet.eu/index.php>) – which is dedicated to collecting information on all newly discovered planets and on the related bibliography – and recently published books (Dvorak, 2008; Seager, 2010; Kasting, 2009; Perryman, 2011).

A great deal of interest is now being paid to the inference, study, and modelling of planetary atmospheres, i.e., in the case of rocky planets “Super-Earths”), the fluid envelope that surrounds the planet and its response to the differential heating due to the absorption of incoming solar radiation. Planetary atmospheres may feature complex chemical compositions and be characterised by the relevant presence of phase transitions of some of these components, which may have large effects on the radiative properties. The geometry describing how the planet and its parent star face each other, determines to a first approximation the horizontal inhomogeneities, whereas the atmospheric optical properties, and in particular its opacity to the incoming radiation, has an effect on the vertical structure (e.g. Earth’s optically thin atmosphere is typically heated from the planets surface). The general circulation of the planetary atmosphere is powered, just like a Carnot engine by such inhomogeneities with, on average, heating occurring in regions of higher temperatures before being then removed in cooler areas. The circulation acts to reduce the temperature gradients through transport processes, and on average the generation of kinetic energy is compensated for by irreversible viscous dissipation processes. Irreversible heat transport and dissipation, as well as, processes such as radiative absorption and emission contribute to the generation of entropy. The steady state is realised by balancing energy and entropy fluxes between the planetary atmosphere and the surrounding space. Various aspects of this closed chain of interdependent processes have been described for terrestrial conditions (Lorenz, 1967; Peixoto and Oort, 1992; Johnson, 2000; Lucarini, 2009; Lucarini et al., 2011), but the theoretical framework proposed in these contributions could be applied with more generality to planetary atmospheres.

The awareness of recent findings in planetary astronomy and astrophysics are gradually filtering through to the classical geophysical sciences (by definition Earth-centred). The study of, e.g. the general circulation of Jupiter, Venus and Mars has a rather old history, but with the recent and foreseeable future discoveries of exoplanets, the opportunity of facing a vast array of planetary configurations, mean moving from considering special cases of climates, to being able to study a quasi-continuous distribution in some parametric space. Following Read (2011), one should note that it is possible to reduce the variety of climate settings by adopting the fluid-dynamical classical method of similarity, i.e. by defining a set of dimensionless numbers that fully characterise the climate state. When two climate states share the same set of dimensionless numbers, they are dynamically equivalent, so that the statistical properties of one can be mapped into those of the other with simple algebraic operations. At this stage, we are left with two additional elements to cope with:

- How to characterise succinctly a climatic state, conveying minimal but comprehensive physical information for planetary atmospheres which are, in general, turbulent fluids with variability on a very vast range of spatial and temporal scales?
- How to verify, or at least approximately, the validity of our simulations and of our theoretical understanding of planetary circulations that we are able to model?

The two items are closely related to the definition of robust observables. Energy conservation imposes that the incoming radiation onto a planet is instantaneously equal to the sum of the radiation that is absorbed and scattered by the planet. Assuming steady state, when averaging over a sufficiently long time interval, the radiation absorbed by the planet is equal to the radiation the planet emits out to space (assuming that the planet has no internal energy sources). These quantities allow defining the average albedo and the effective thermodynamic temperature, which constitute the most fundamental description of the properties of a

planet, in terms of the first law of thermodynamics. As recorded by a telescope, observations made of a planet represented by a single pixel, provides a sufficient level of detail to gather such information. It is then possible for the overall macroscopic properties of the planets atmosphere to be found. Obviously, most observations provide much more than this, such as the emission spectrum of a planet (and that of the star in orbits). Moreover, spectral windows may prove useful for measuring its surface temperature. Nonetheless, in terms of the macroscopic thermodynamic properties of the planet, the spectrally integrated quantities are the most relevant.

Under conditions of steady state, given the inhomogeneity of the incoming radiation and of the boundary conditions, the vanishing net energy budget at the top of the atmosphere (TOA) results from the cancellation between regions of net heating and regions of net cooling (low and mid-high latitudes regions, in the case of the Earth). Steady state arguments imply that the absolute value of the imbalance in either region is equal to the energy transmitted from the regions where the energy balance at the top of the atmosphere is positive from those where such balance is negative, and the transport is performed across the atmospheric fluid envelope through material transport and horizontal deflection of the incoming radiation (in the case of very thick atmospheres). Note that the correct representation of these large scale properties in state-of-the-art models of the Earth’s climate is a far from trivial task (Lucarini and Ragone, 2011).

Moreover, the second law of thermodynamics imposes that the regions featuring positive TOA energy budget are warmer (i.e. their total emitted radiation is larger) than those where the balance is negative. One can show that the entropy production due to such irreversible transport of energy from hot to cold regions gives a lower bound to the total material entropy production of the planet. Moreover, it is possible to provide a lower bound to the total dissipation of kinetic energy or, equivalently, to the intensity of the Lorenz energy cycle (Lucarini et al., 2011). This discussion implies that if a telescope allows for an observational resolution involving more than one pixel, and is able to distinguish between warmer and colder regions, it is possible to derive fundamental information on the macroscopic thermodynamic properties of the planet in terms of the 2nd law of thermodynamics. This brief discussion suggests that non-equilibrium thermodynamics provides rather powerful methods and concepts for analysing the fundamental properties of planetary atmospheres, as well as a means of testing model performance, and deriving bounds for the physical properties of the system even when low-resolution data is only available. Following previous work in the field (Pujol, 2003; Fraedrich and Lunkeit, 2008; Lucarini, 2009; Lucarini et al., 2010b, 2011; Pascale et al., 2011a,b; Li and Chylek, 2012), we propose using this framework in order to study planetary atmospheres in a rather general setting of forcings and boundary conditions.

1.2. Habitability conditions and climatic bistability

As mentioned above, a great deal of interest in the investigation of Super-Earth planetary atmospheres is directed at studying conditions within or around the Habitable zone. Nonetheless, being in the Habitable zone is a *necessary but not a sufficient condition* for a planet to have liquid water at the surface, even if the chemical composition of the atmosphere would in principle allow for it. In fact, the paleohistory of our planet provides strong evidence that the astronomical and astrophysical parameters of the Sun–Earth system support two distinct steady states: a warm state (W) characterised by widespread liquid water and a moist atmosphere; with the other characterised by the global glaciation of water and an extremely dry atmosphere, i.e. the so-called Snowball state (SB). Proposed here is a thorough investigation of these two states in a bi-dimensional parameter space, with the goal of detailing the

region of bi-stability for an Earth-like system in order to analyse the bifurcation points and the mechanisms of transitions between both states. Our investigation deals with Earth-centric conditions because this is the only system where an extensive body of literature exists for the Snowball state and the Snowball-Snowfree transitions, be it observational, theoretical, or model-assisted, however, the scope of applicability may be much wider.

Probably the most notable examples of climate change events on the Earth occurred during the Neoproterozoic (period spanning from 1000 to 540 myr ago), when the Earth is believed to have suffered two of its most severe periods of glaciation (Hoffman et al., 1998) and entered into a SB climate state. This period coincided with large carbon dioxide fluctuations, while the solar constant (about 1360 W m^{-2} under present conditions) is believed to have been 94% of current levels, rising to 95% by the end of the Neoproterozoic era (Gough, 1981; Pierrehumbert et al., 2011). The two main factors effecting the concentration of atmospheric CO_2 are biotic activity and volcanism. Volcanic eruptions bring about very sudden and dramatic increases in CO_2 concentration. This process provides a potential mechanism through which the climate state can exit the SB condition, by increasing the opacity of the atmosphere and enhancing the greenhouse effect. The biospheric effect tends to occur more gradually as the biotic activity and atmospheric composition are coupled so that large fluctuations of the carbon pools take place over long time scales. Note that the effect of SB events on the biosphere is believed to have been disastrous. Carbon-isotope ratios characteristic of Earth's mantle (Hoffman et al., 1998; Kennedy et al., 1998) rather than of life processes recorded immediately below and above the glacial deposits imply that oceanic photosynthesis was effectively non-existent during SB events. The result of this and anoxic conditions beneath the ice should have led to the disappearance of most kinds of forms of life except bacteria. The final disappearance of SB conditions since the Neoproterozoic may have been the main contributing factor in the development of complex multi-cellular life that began around 565 myr ago.

Based on the evidence supported by Hoffman et al. (1998) and Hoffman and Schrag (2002), it is therefore expected that the Earth is potentially capable of supporting multiple steady states for the same values of some parameters such as the solar constant and the concentration of carbon dioxide, which directly affect the radiative forcing. It is important therefore to explore this hypothesis, due to its relevance for the history of our planet but also to help understand other planets capabilities for supporting life.

Initial research using simple 0-D models (Budyko, 1969; Sellers, 1969), 1-D models (Ghil, 1976) as well as more recent analyses performed using complex 3-D general circulation models (Marotzke and Botzet, 2007; Voigt and Marotzke, 2011; Pierrehumbert et al., 2011), provide support for the existence of such bi-stability. The SB $\rightarrow$ W and W $\rightarrow$ SB transitions tend to occur as an abrupt rather than a smooth transition. The main mechanism triggering such abrupt transitions is the positive ice-albedo feedback (Budyko, 1969; Sellers, 1969). Such a feedback is associated with the fact that as global temperatures increase, the extent of snow and ice cover decreases thus reducing the albedo and therefore increasing the amount of sunlight absorbed by the Earth system. Conversely, a negative fluctuation in the temperature leads to an increase in the albedo therefore reinforcing the cooling.

The presence of such catastrophic climate shifts (Arnold, 1992) suggest the existence of a global bifurcation in the climate system for certain combinations of its descriptive parameters (Fraedrich, 1979). The loss of stability realised in the W $\rightarrow$ SB and SB $\rightarrow$ W transitions is related to the catastrophic disappearance of one of the two attractors describing the two possible climatic states, as a result of a set of complex bifurcations.

Starting from present conditions, the most obvious physical parameters to modulate in order to bring about the transition to

the SB state is the solar constant. Even if other model experiments (Voigt and Marotzke, 2011) show that a decrease in CO_2 alone can bring about the transition to the SB state, this requires more than an 80% decrease in CO_2 concentration, compared to a decrease of less than 10% for the solar luminosity. The Neoproterozoic however highlights the importance of considering changes in CO_2 levels as a mechanism for transition to and from the SB climate state, and the dramatic impact it can have on the overall state of the climate system. It is therefore interesting to alter both the solar luminosity and the atmospheric opacity as these are two important parameters affecting the overall properties of the system. If one wants to explore extensively a parameter space of climate steady states in terms of these two quantities, it is therefore necessary to consider a wide range of values.

Using PlaSim, a general circulation model of intermediate complexity (Fraedrich et al., 2005), we study the climate states realised when the solar constant is modulated between 1160 W m^{-2} and 1510 W m^{-2} and the values of $[\text{CO}_2]$ are varied between 90 and 2880 ppm. Our aim here is an empirical (i.e. based on model simulations) reconstruction of the global structural properties of the climatic attractors. The work proposed here builds on results shown in Lucarini et al. (2010a,b), where only one of the two climatic parameters described above was varied. For both W and SB states we compute surface temperature, material entropy production, meridional energy transport, Carnot efficiency and intensity of the Lorenz energy cycle, using the theoretical framework proposed in Johnson (2000) and Lucarini (2009). We will look for an empirical relation for the two transition lines (W $\rightarrow$ SB and SB $\rightarrow$ W) in the parametric plane between S^* and the natural logarithm of $[\text{CO}_2]$ which marks the boundaries of the hysteresis in the climate system. The aforementioned thermodynamic quantities are used to explain changes in large-scale climate behaviour and the effect of climate change on features such as stratification and baroclinicity in order to understand changes in the meridional heat transport across the parameter range. It will also be shown that the Carnot efficiency has a key role in defining the stability of the system which is related to abrupt climatic shifts. Moreover, we propose empirical functions allowing for expressing the global non-equilibrium thermodynamical properties of the system in terms of either the mean surface temperature or the mean planetary emission temperature. This is indeed promising, because it suggests the possibility of expressing thermodynamic properties which are very hard to observe experimentally, as functions of quantities which can be measured.

Note, our work differs from most others done on the Snowball state, in the sense that we explore the Snowball state using the Earth's present land configuration as opposed to how it was during the Neoproterozoic era. It is well known that the land mass distribution is important in determining the specific details of the transitions between the SB and W states (Voigt et al., 2011). Nonetheless we do not expect that the structural properties of the system will significantly be affected when a wide range of values of the solar constant and CO_2 concentration are considered.

Another aspect to mention in regard to interpreting our results for more general planetary systems, is their sensitivity to the spectral distribution of stellar irradiance and atmospheric composition. Obviously, a different spectrum of incoming stellar radiation and optical properties of the atmosphere will surely change the results. However, so long as radiation is mostly absorbed at the surface; clouds have the dual effect of scattering shortwave and trapping long wave radiation; ice, most importantly, has a much higher albedo than land or ocean; and we have a greenhouse gas that can be parametrically modulated (that is equivalent to CO_2), the basic properties described here will obviously be affected but not altered in a fundamental way.

Note, we do not expect a priori in Paleoclimatic terms, nor universal relations for planetary systems. However, we do propose

new potentially relevant methods for addressing both problems. In doing so, we hope to move a small step in the direction of deriving more general properties, by showing that reducing the rotation rate of the planet by a factor of 2 does not alter significantly any of the properties obtained with the terrestrial value of the rotation rate, despite specific meteorological features of the planet being very different. Of course, such sensitivity analysis is not exhaustive and needs to be completed in future studies by looking at the impact of changing other astronomical and astrophysical parameters.

The paper is structured in the following way: in Section 2 we describe details of the non-equilibrium thermodynamics of the climate and the diagnostic tools used. Section 3 is dedicated to the description of the PlaSim climate model and of the experimental setup. In Section 4 we discuss the results of the simulations, in Section 5 we propose parameterisations of the main non-equilibrium properties of the system as a function of the mean surface temperature or emission temperature. Further to this, in Section 6 we discuss the effect of reducing the rotation rate of the planet by a factor of 2, and in Section 7 we present our conclusions.

2. Non-equilibrium thermodynamics of the climate

In this section we recapitulate some thermodynamic properties of the climate system and introduce the notation used throughout this chapter. For sake of clarity and completeness, we summarise here what has been presented in greater detail in Lucarini (2009). If the climate system is encompassed by a domain V , the total energy budget is given by $E(V) = P(V) + K(V)$, where K represents the total kinetic energy and P is the moist static potential energy, which includes contributions from the thermal (including latent heat) and potential energy. The time derivative of K and P can be found to be $\dot{K} = -D + W$ and $\dot{P} = \dot{\Psi} + D - W$, where $D = \int_V dV \rho \epsilon^2$, is the dissipation and therefore always positive, W is the instantaneous work done by the system and $\dot{\Psi} = \int_V dV (-\nabla \cdot H)$, is the total heating due to convergence of turbulent heat fluxes and radiative fluxes, such that $\dot{E} = \dot{\Psi}$. The dependence on V has been dropped for convenience. As soon as the climate is considered as a non-equilibrium steady state system (NESS, see Gallavotti (2006)), we have that over long time scales $\bar{\dot{E}} = \bar{\dot{P}} = \bar{\dot{K}} = 0$ (the bar indicates averaging over long time periods). Let us define $\dot{Q}$ as the local heating rate so that $\dot{Q} = \rho \epsilon^2 - \nabla \cdot H$ (Lucarini, 2009) where $\epsilon^2 > 0$ is the local rate of heating due to viscous dissipation of kinetic energy and H is given by the sum of turbulent heat fluxes plus radiative fluxes. We split the domain V into the subdomain V^+ , where $\dot{Q} = \dot{Q}^+ > 0$, and V^- , where $\dot{Q} = \dot{Q}^- < 0$. We obtain:

$$\dot{\Psi} + D = \dot{P} + W = \int_{V^+} dV \rho Q^+ + \int_{V^-} dV \rho Q^- = \dot{\Phi}^+ + \dot{\Phi}^- = \dot{\Phi} \quad (1)$$

Where the quantities $\dot{\Phi}^+$ and $\dot{\Phi}^-$ are positive and negative at all times, respectively. Since dissipation is positive definite, $-\bar{\dot{K}} + \bar{\dot{W}} = \bar{\dot{D}} = \bar{\dot{P}} + \bar{\dot{W}} = \bar{\dot{W}} = \bar{\dot{\Phi}^+} + \bar{\dot{\Phi}^-} > 0$. A diagram showing the energy conversions and reservoirs is given in Fig. 1.

On spatial scales far smaller than V itself, it is practical to assume local equilibrium DeGroot and Mazur (1984) so that locally $\dot{Q} = \dot{s}T$ with $\dot{s}$ the time derivative of the entropy density. The total rate of change of the entropy of the system is:

$$\begin{aligned} \dot{S} &= \int_{V^+} dV \rho \dot{Q}^+ / T + \int_{V^-} dV \rho \dot{Q}^- / T \\ &= \int_{V^+} dV \rho |\dot{s}^+| + \int_{V^-} dV \rho |\dot{s}^-| = \dot{S}^+ + \dot{S}^- \end{aligned} \quad (2)$$

where $\dot{S}^+ > 0$ and $\dot{S}^- < 0$. Using Eq. (2) and assuming that the Earth system is in a steady state, over a long time average, $\bar{\dot{S}}^+ = -\bar{\dot{S}}^-$ as $\bar{\dot{S}} = 0$. Therefore, $2\bar{\dot{S}}^+ = \int_V dV \rho |\dot{s}|$, so that $\bar{\dot{S}}^+$ measures the absolute value of the entropy fluctuations throughout the domain. When

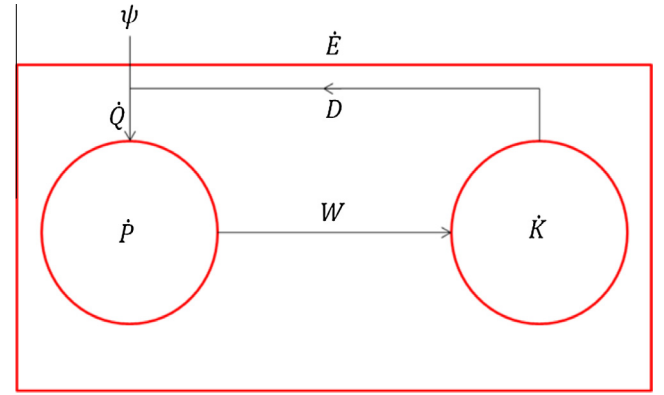


Fig. 1. Schematic figure showing the two main classes of energy: Potential (P) and Kinetic (K); and the transfer processes (ψ, W, D) which exchange energy between them. $\dot{E}$ is the total energy within the system and $\dot{Q} = \psi + D$ is the total heating rate due to dissipation (D) and the convergence of heat fluxes (ψ). W is the work done by the system. ψ includes the external radiative input to the climate system from the sun, therefore $\dot{E} = \dot{\psi}$.

integrating over the whole domain and considering long time average, we have the following equivalent expressions for the thermodynamic quantities: $\bar{\dot{\Phi}}^+ = \bar{\dot{S}}^+ \bar{\Theta}^+$ and $\bar{\dot{\Phi}}^- = \bar{\dot{S}}^- \bar{\Theta}^-$, where $\bar{\Theta}^+$ and $\bar{\Theta}^-$ are the time and space averaged temperatures of the V^+ and V^- domains respectively. Since $|\bar{\dot{S}}^+| = |\bar{\dot{S}}^-|$ and $|\bar{\Phi}^+| > |\bar{\Phi}^-|$, it can be shown that $\bar{\Theta}^+ > \bar{\Theta}^-$, i.e. absorption typically occurs at higher temperature than release of heat (Peixoto et al., 1991; Johnson, 1997). The work done by the Carnot engine of the climate system is found to be, $\bar{W} = \eta \bar{\dot{\Phi}}^+$, where

$$\eta = \frac{\bar{\dot{\Phi}}^+ + \bar{\dot{\Phi}}^-}{\bar{\dot{\Phi}}^+} \quad (3)$$

can be defined as the Carnot efficiency of the system.

As shown in Lorenz (1967) and clarified in Johnson (1997, 2000) the long term average of the work performed by the system is equal to the long-term average of the generation of available potential energy, as is typical of forced-dissipative steady states. The Earth exists in a steady state maintained far from equilibrium by net radiative heating in the warm region (low latitudes on Earth) and net cooling at the cold regions (high latitudes on Earth), which are compensated by large-scale transports performed by the planetary atmosphere. Note that η provides a clear measure of how far the system is from equilibrium. The entropy production due to the irreversibility of processes occurring within the climatic fluid is called the material entropy production, $\dot{S}_{mat}$ and can be written in general terms as:

$$\dot{S}_{mat} = \int_V \frac{\epsilon^2}{T} dV + \int_V \vec{F}_{SH} \cdot \nabla \frac{1}{T} dV + \int_V \vec{F}_{LH} \cdot \nabla \frac{1}{T} dV \quad (4)$$

where the first, second, and third terms on the RHS are related to the dissipation of kinetic energy, and to the transport of sensible and latent heat across temperature gradients, respectively. We now link the terms of the entropy budget in Eq. (2) with those of the entropy production in Eq. (4). The second law of thermodynamics states that the entropy variation of a system at temperature T receiving an amount of heat δQ is larger than or of equal to $\delta Q/T$ Landau and Lifshitz (1980). In this case:

$$\begin{aligned} \bar{\dot{S}_{mat}}(V) &\geq \bar{\dot{S}_{min}}(V) = \left(\frac{\int_V dV \rho \dot{Q}}{\int_V dV \rho T} \right) = \left(\frac{\bar{\dot{\Phi}}^+ + \bar{\dot{\Phi}}^-}{\langle \Theta \rangle} \right) \approx \frac{\bar{\dot{\Phi}}^+ + \bar{\dot{\Phi}}^-}{\langle \Theta \rangle} \\ &\approx \frac{\bar{W}}{\langle \Theta \rangle} \approx \frac{\bar{W}}{\frac{\langle \Theta^+ + \Theta^- \rangle}{2}} \end{aligned} \quad (5)$$

where $\overline{\dot{S}_{mat}(V)}$ is the long-term average of the material entropy production, $\dot{S}_{min}(V)$ is its lower bound, i.e. the minimal value of the entropy production compatible with the presence of a Lorenz energy cycle with average intensity $\overline{W}$ and $\langle\Theta\rangle$ is the density averaged temperature of the system. The approximation holds as long as we can neglect the impact of the cross-correlation between the fluctuations of the work and the fluctuations of the globally averaged temperature. This term is small because the fluctuation of the globally averaged temperature are determined by the fluctuations of the radiative balance, which are weakly correlated to those of the work.

Moreover, in the last approximation we assume that $\langle\Theta\rangle$ can be estimated as the mean of the two Carnot temperatures Θ^+ and Θ^- . This can be justified by assuming that the atmospheric volumes where net heating and net cooling occur are of comparable size. This is indeed true on Earth, where the regions of net atmospheric warming are well approximated to be centred around the Equator between about [30°S, 30°N], with all other regions associated with net cooling (Lucarini et al., 2010a; Donohoe and Battisti, 2012).

We can explicitly write $\dot{S}_{min}(V)$ as:

$$\begin{aligned}\dot{S}_{min}(V) &\approx \frac{\overline{W}}{(\Theta^+ + \Theta^-)/2} = \frac{\eta \overline{\Phi^+}}{(\Theta^+ + \Theta^-)/2} = \eta \frac{\Theta^+}{(\Theta^+ + \Theta^-)/2} \overline{\dot{S}^+} \\ &= \frac{\eta}{1 - \eta/2} \overline{\dot{S}^+} \approx \eta \overline{\dot{S}^+}\end{aligned}\quad (6)$$

where the last approximation holds as long as $\eta \ll 1$, which is typically the case. Therefore, η sets also the proportionality factor relating the lower bound to the entropy production of the system $\dot{S}_{min}(V)$ due to macroscopically irreversible processes to the absolute value of the entropy fluctuations inside the system due to macroscopically reversible heating or cooling processes. Note that if the system is close to equilibrium ($\eta \rightarrow 0$), the entropy production vanishes because there is no generation of kinetic energy and there is no transport of heat. The lower bound to the material entropy production corresponds to the contribution coming from the dissipation of kinetic energy through viscous processes. Therefore, the average material entropy production can be expressed as $\dot{S}_{mat} = \dot{S}_{min} + \dot{S}_{exc}$, where $\dot{S}_{exc}$ is the excess of entropy production with respect to the minimum, which results from the heat transport down the temperature gradient (Lucarini, 2009). We can define:

$$\alpha \approx \frac{\overline{\dot{S}_{exc}}}{\dot{S}_{min}} \approx \frac{\int_V dV (\overline{\vec{F}_{SH}} + \overline{\vec{F}_{LH}}) \cdot \nabla (\frac{1}{T})}{(\overline{W}/\langle\Theta\rangle)} \geq 0 \quad (7)$$

as a parameter of the irreversibility of the system, which is zero if all the production of entropy is due to the unavoidable viscous dissipation of the mechanical energy. As $\dot{S}_{mat} \approx \eta \overline{\dot{S}^+} (1 + \alpha)$, we have that the entropy production is maximised if we have a joint optimisation of heat transport and the production of mechanical work. Note that, if heat transport down the temperature gradient is very strong, the efficiency η is small because the difference between the temperatures of the warm and cold reservoirs is greatly reduced (the system is almost isothermal), whereas, if the transport is very weak, the factor α is small. The parameter α introduced above is related to the Bejan number Be as $Be = \alpha + 1$ (Paoletti et al., 1989).

3. Experimental setup

We study dynamical and thermodynamical properties of the Earth's climate using the PlaSim climate model. This is motivated by the fact that our study requires altering very extensively some parameters of the climate system and producing many simulations. Therefore we need a model which is flexible and fast to run rather than a state-of-the-art model encompassing as many processes occurring in the Earth as possible.

PlaSim (Fraedrich et al., 2005) is a climate model of intermediate complexity, freely available at <http://www.mi.uni-hamburg.de/plasim>. Its dynamical core is formulated using the primitive equations for vorticity, divergence, temperature, specific humidity, and the logarithm of surface pressure, solved using the spectral transform method (Eliassen et al., 1970; Orszag, 1970) with semi-Lagrangian advection. Unresolved processes for long (Sasamori, 1968) and short (Lacis and Hansen, 1974) wave radiation, shallow, moist (Kuo, 1965, 1974) and dry convection, cloud formation (Stephens, 1978; Stephens et al., 1982; Slingo and Slingo, 1991) and large scale precipitation, latent and sensible heat boundary layer fluxes, horizontal and vertical diffusion (Louis, 1979; Louis et al., 1981; Laursen and Eliassen, 1989) are parameterized. The model is coupled to a 50-m deep mixed layer ocean which contains a thermodynamic sea-ice model. The advantage of using a slab ocean as opposed to a full ocean is that it allows for the climate system to reach a steady state in less than 35 years after a change in e.g. the solar constant. With full ocean coupling, the integration time of the model and the time needed to reach a steady state would be an order of magnitude larger (Voigt and Marotzke, 2011). We wish to emphasise that whereas most state-of-the-art general circulation models feature an imperfect closure of the global energy budget in the order of 1 W m^{-2} for standard climate conditions, due to inaccuracies in the treatment of the dissipation of kinetic energy and the hydrological cycle (Lucarini and Ragone, 2011; Liepert and Previdi, 2012). In our simulations, which cover very extreme climate conditions, the energy bias is always smaller than 0.5 W m^{-2} . Therefore, we are confident of the thermodynamic consistency of our model. Moreover, a tested entropy diagnostic is available (Fraedrich and Lunkeit, 2008), thus making it well suited for this work.

The model is run at T21 resolution (approximately $5.6^\circ \times 5.6^\circ$) with 10 vertical levels. While this resolution is relatively low, it is expected to be sufficient for obtaining a reasonable description of the large scale properties of the atmospheric dynamics, which are most relevant for the global features we are interested in. We remark that previous analyses have shown that using a spatial resolution approximately equivalent to T21 allows for obtaining an accurate representation of the major large scale features of the climate system and of its global thermodynamical properties (Jones et al., 2005; Pascale et al., 2011a). Moreover, since we perform an extensive parametric investigation, we need a parsimonious climate model.

As already discussed in Lucarini et al. (2010b), this model features multistability for $[\text{CO}_2] = 360 \text{ ppm}$: there is a range of values of S^* centred approximately around the present value of $S^* = 1360 - \text{W m}^{-2}$ such that for each choice of S^* , there are two attractors, and so two distinct climates, exist. One climate is characterised by extremely cold and dry conditions, and we refer to it as the Snowball (SB) state. The other climate, instead, resembles present-day conditions and we refer to this as the Warm (W) state. For values of S^* larger than those included in the range of multistability, only the W state is present, whereas only the SB state is present for low values of S^* . The two critical values of S^* (Fig. 2) determining the range of multistability are the tipping points of the system. In this paper, we wish to extend this analysis for different values of $[\text{CO}_2]$, in order to treat in more generality the problem of multistability of the climate system. Changing S^* over a wide range of $[\text{CO}_2]$ of 90, 180, 270, 360, 540, 720, 1080, 1440, 2160 and 2880 ppm, we are able to reconstruct the SB and W climate states. The procedure occurs as follows for each of the considered values of $[\text{CO}_2]$:

1. the model is run to a W steady state for 100 years with S^* equal to 1510 W m^{-2} ;
2. S^* is decreased by a small amount for each value of CO_2 and the model run is continued until a steady state is reached;

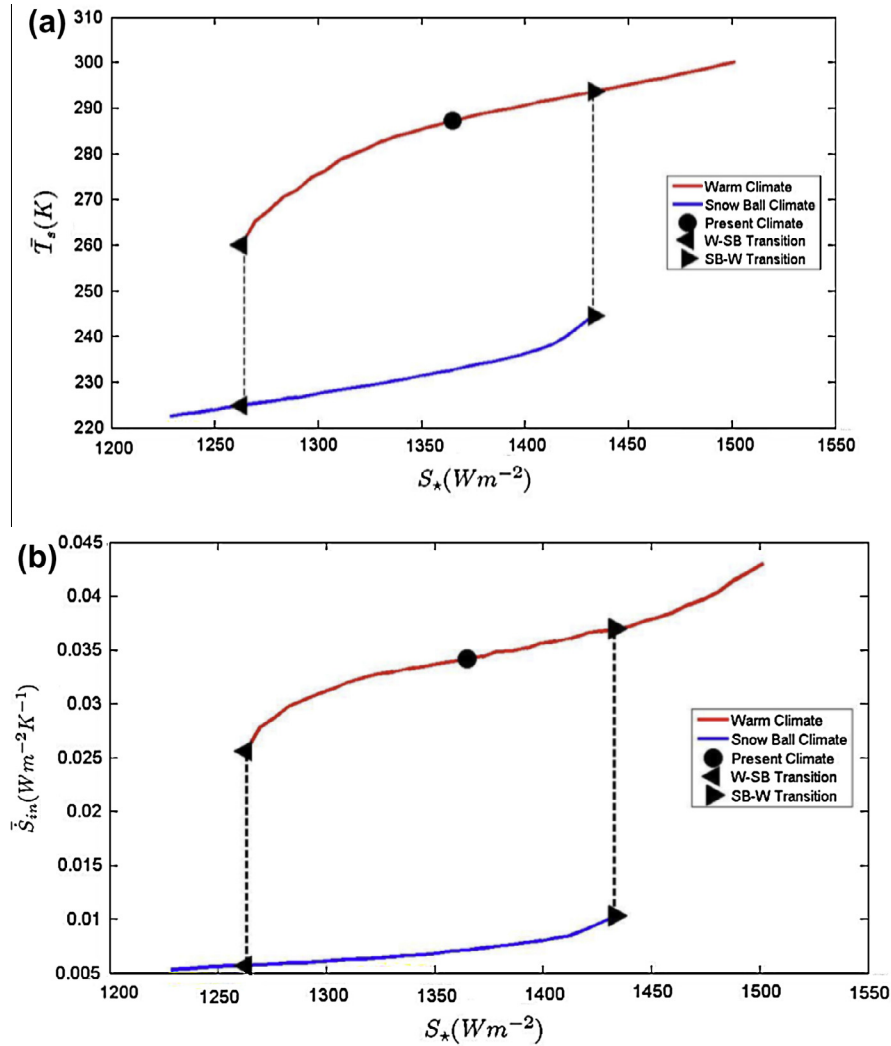


Fig. 2. (a) Surface temperature and (b) material entropy production for steady states obtained for different values of the solar constant S^* . The present climate is marked with a black circle, the W states in red and the SB states in blue. Adapted from Lucarini et al. (2010b). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

3. step 2 is repeated until S^* is reduced to 1165 W m^{-2} ; the point of $W \rightarrow \text{SB}$ transition is noted down;
4. the reverse operation is then performed with S^* increased step by step, up to the value of 1510 W m^{-2} , each time allowing the system to reach a steady state; the point of $\text{SB} \rightarrow W$ transition is noted down.

Further to this, we identify the position of the transition to a higher resolution than the rest of the parameter range in the direction of S^* . For values of S^* within 10 W m^{-2} before the transition, S^* is decreased in intervals of 1 W m^{-2} , each time permitting 50 years for the system to reach a steady state, until after the transition is observed.

We wish to address a caveat related to the appropriateness of the model used in this study. The radiation scheme adopted by PLASIM has been devised to deal with Earth-like CO_2 concentrations, and in this range it is reasonably accurate. While in this study we do push the radiation code beyond the conditions usually considered, we are still using CO_2 concentrations which are within one order of magnitude of those found currently and therefore, retain confidence in our model. Moreover, this study does not intend to obtain precise quantitative estimates as a result of strong perturbations to the atmospheric opacity and/or solar irradiance, but

rather at deriving the overall behaviour of the system. The effect of considering ultra-high values of CO_2 concentrations (much higher than what has been considered here) has been analysed in detail by Pierrehumbert (2005), who discussed how using a radiative model designed for Earth climate conditions can be problematic. Note that for very high CO_2 concentrations additional effects may become relevant like the onset of CO_2 condensation, which is not described by the microscopic physics of climate models.

4. Hysteresis, bistability and regime boundaries in a parametric space

Initially, focus is put on analysing the parametric plane ($[\text{CO}_2]$, S^*), which shall be referred to as the CS space, as a function of global mean surface temperature, T_s . The transition zones between the climate states are clearly defined from the temperature profiles. Note, the two qualitatively distinct properties of the climate system in the CS space, namely the presence of bi-stability, can be reconstructed from any observable of the climate state, but it is most instructive to select first the surface temperature. The temperature profile through the CS space is illustrated in Fig. 3a and b.

We can identify two main climatic regimes, observed as two distinct manifolds ($[\text{CO}_2]$, S^* , T_s) and characterised by a sharp

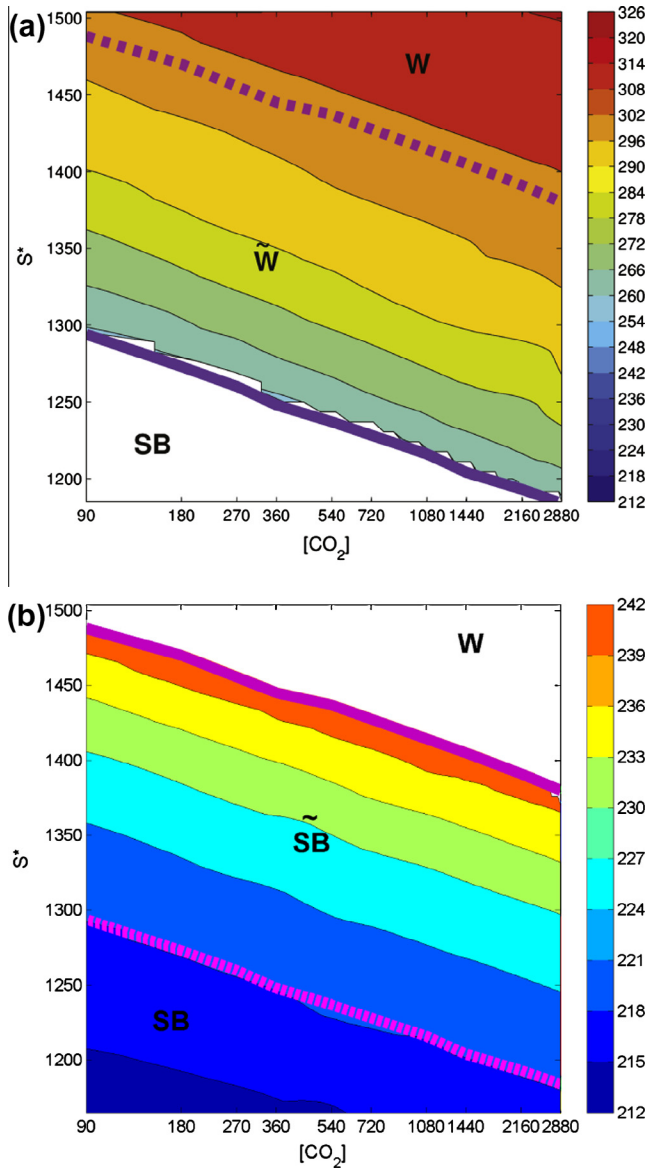


Fig. 3. Contour plot of surface temperature (K) as a function of S^* and the $[\text{CO}_2]$. The lower SB (a) and upper W (b) manifolds are shown. The transition $\text{SB} \rightarrow \text{W}$ and $\text{W} \rightarrow \text{SB}$ are shown by the upper and lower purple lines respectively. The blue dots indicate the values of $(S^*, [\text{CO}_2])$ for which simulations have been performed. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

change in T_s when jumping from one manifold to another. We refer to these as the upper and lower manifolds, representative of the W and SB regimes respectively. As expected, there is a monotonic increase of temperature with increasing CO_2 or S^* on both manifolds (Voigt and Marotzke, 2011; Pierrehumbert, 2005). Across the explored parametric space, the temperature range on the SB and W manifolds are 212–242 K, and 254–326 K respectively. We see that the temperature range 242–254 K is not permitted by the climate system.

The temperature range of the bistable region in the SB and W regimes are 218–242 K and 254–300 K respectively, meaning that the rate of change of surface temperature over the same range of S^* and $[\text{CO}_2]$ is approximately twice as large in the W regime with respect to the SB regime. The climate sensitivity of the W state is much higher than that of the SB state because SB is almost entirely dry, so that the positive water vapour feedback is not active and the surface temperature difference between the two manifolds

ranges between 40 K and 60 K, respectively, for identical values of S^* and $[\text{CO}_2]$. The W states (upper manifold) exists only in the region of the CS space above the $\text{W} \rightarrow \text{SB}$ transition line (the position of this line is expressed as $S^* = S_{sbw}^*$) whereas the SB states (lower manifold) only in the CS region below the $\text{SB} \rightarrow \text{W}$ transition line (the position of this line is expressed as $S^* = S_{wsb}^*$). Such lines, which are well separated and approximately parallel, are illustrated as solid and dashed purple dashed lines in Figs. 3a–9b and have been found within an accuracy of 2 W m^{-2} of the solar constant. The solid purple lines indicate the tipping points: for the SB states they correspond to the $\text{SB} \rightarrow \text{W}$ transitions, for the W states they correspond to the $\text{W} \rightarrow \text{SB}$ transitions, respectively. Instead, when considering the W (SB) manifold, the dashed lines indicate the location of the tipping points destabilising the SB (W) states. The bistable region is therefore located between the dashed and solid purple lines. As a result, a property of the system is that regardless of which combination of $[\text{CO}_2]$ and S^* is used, the transition from one state to another always occurs at almost exactly the same temperature. This indicates that the climate system is insensitive to the mechanism of forcing. The position of the two boundaries can be parameterised in terms of S^* and CO_2 concentration as:

$$S_{sbw}^* = a_{sbw} \log_{10}[\text{CO}_2] + C_{sbw}, \quad S_{wsb}^* = a_{wsb} \log_{10}[\text{CO}_2] + C_{wsb} \quad (8)$$

where $a_{sbw} \approx a_{wsb} \approx -70 \text{ W m}^{-2}$, $C_{sbw} \approx 1630 \text{ W m}^{-2}$ and $C_{wsb} \approx 1440 \text{ W m}^{-2}$ for the transition $\text{SB} \rightarrow \text{W}$ and $\text{W} \rightarrow \text{SB}$ respectively and $[\text{CO}_2]$ is expressed in ppm. The size of the bistable region, which we define as B , along S^* can therefore be defined by the difference between C_{sbw} and C_{wsb} :

$$B = C_{sbw} - C_{wsb}. \quad (9)$$

It is found that B is approximately 200 W m^{-2} . The displacement between the position of the boundaries gives a precise measure of the hysteretic properties of the climate (Budyko, 1969; Sellers, 1969; Voigt and Marotzke, 2011; Lucarini et al., 2010b) since it indicates the size of the overlap between the two manifolds in the CS plane. The presence of a bistable region implies that when we change the values of S^* and $[\text{CO}_2]$ from an initial to a final value, the final steady state depends on the initial steady state and on the path of change of S^* and $[\text{CO}_2]$. Let us assume that we start from an initial point $(S_0^*, [\text{CO}_2]_0)$ in the bistable region and in the W state. Let us also assume we perform a close path of variation of S^* and $[\text{CO}_2]$ so that $S_0^* = S_f^*$ and $[\text{CO}_2]_0 = [\text{CO}_2]_f$. If the path does not cross S_{wsb}^* the final state will be identical to the initial one, that is, in a time-averaged sense:

$$T_s(S_0^*, [\text{CO}_2]_0) = T_s(S_f^*, [\text{CO}_2]_f). \quad (10)$$

On the other hand, if the closed path crosses the transition line to the second manifold, the final state will be different from the initial:

$$T_s(S_0^*, [\text{CO}_2]_0) \neq T_s(S_f^*, [\text{CO}_2]_f). \quad (11)$$

If, furthermore, the closed path crosses first S_{wsb}^* and then S_{sbw}^* , then again $T_s(S_0^*, [\text{CO}_2]_0) = T_s(S_f^*, [\text{CO}_2]_f)$, since the system has performed first a $\text{W} \rightarrow \text{SB}$, and then a $\text{SB} \rightarrow \text{W}$ transition. The same applies starting from a SB state and exchanging SB with W in the previous discussion. This is true for any climate diagnostic. More specifically, in the case of T_s , for $\text{W} \rightarrow \text{SB}$ and $\text{SB} \rightarrow \text{W}$ transitions,

$$T_s(S_0^*, [\text{CO}_2]_0) > T_s(S_f^*, [\text{CO}_2]_f) \quad \text{and} \quad T_s(S_0^*, [\text{CO}_2]_0) < T_s(S_f^*, [\text{CO}_2]_f) \quad (12)$$

respectively. Note that for $S^* > 1440 \text{ W m}^{-2}$, even if $[\text{CO}_2]$ is 0 ppm, no transition to SB state can occur. Furthermore, it is important to underline that while it is reasonable to expect that some details of the characterisations of the phase portrait of the system given

above may depend on our model and on its parameterisations, the qualitative features are expected to remain robust.

Fig. 4a and b shows the reconstruction for the material entropy production, $\bar{S}_{mat}$ in CS space, computed directly as described in Fraedrich and Lunkeit (2008). As with temperature, $\bar{S}_{mat}$ increases monotonically with increasing S^* and $[CO_2]$ on both manifolds. In the SB state, the entropy is mostly generated by dissipation of kinetic energy and by irreversible sensible heat transport, because the planet is almost entirely dry. For the W manifold the main contribution to entropy production comes from latent heat due to large-scale and convective precipitation. In the bistable region, the range of $\bar{S}_{mat}$ is $(10, 19) \cdot 10^{-3} \text{ W m}^{-2} \text{ K}^{-1}$ and $(34, 62) \cdot 10^{-3} \text{ W m}^{-2} \text{ K}^{-1}$ for the SB and W respectively, therefore a factor of 3 larger in the W regime with respect to the SB regime. This confirms that $\bar{S}_{mat}$ may be a better indicator than temperature for discriminating the SB and W states as already discussed in Lucarini et al. (2010b). Again, there is a range of values of $\bar{S}_{mat}$ – from 19 to $34 \cdot 10^{-3} \text{ W m}^{-2} \text{ K}^{-1}$ – which is not allowed by the system.

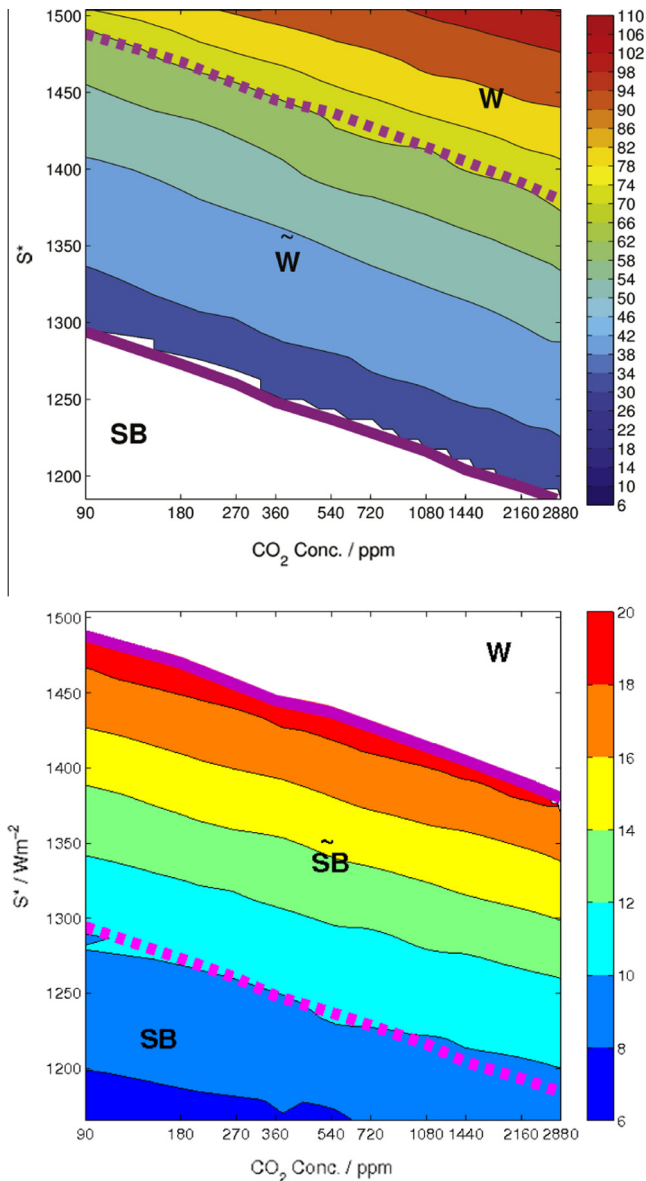


Fig. 4. Contour plot of $\bar{S}_{mat}$ ($10^{-3} \text{ W m}^{-2} \text{ K}^{-1}$) as a function of S^* and the $[CO_2]$ for the lower SB (a) and upper W (b) manifolds.

In the bistable region, the SB and W states are quantitatively very different with respect to their physical properties. The two disjoint attractors can be thus thought of as representing two different worlds, with completely different dynamical and thermodynamical properties. For this reason in the following two subsections dynamical and thermodynamical properties of the manifolds will be analysed individually in terms of the vertical and horizontal surface temperature differences, Carnot efficiency, meridional heat transport and dissipation of kinetic energy. Furthermore we shall relate these properties to the average mean global temperature and the material entropy production. As is illustrated as solid and dashed purple lines in the CS space figures, each manifold will be divided up into two sub regions: W, W/Bistable, on the upper manifold and SB, SB/Bistable on the lower. Then, in a third sub-section, we will analyse the transitions between the two manifolds occurring in bifurcations regions.

4.1. The warm state

The meridional energy transport profile is worked out (Peixoto and Oort, 1992; Lucarini et al., 2011) by integrating over latitudes, the longitudinally averaged TOA radiation budget. We then define as a scalar indicator of the transport, half of the sum of the peak values of the poleward heat transport in both hemispheres, MET . In a moist atmosphere, the average global temperature and the meridional surface temperature difference, defined in our case as the difference between the mean surface temperature of the tropical (30S, 30N) and the polar (90S, 60S) and (60N, 90N) regions (see Fig. 6a), are the main contributing factors for controlling the meridional heat transport (see Fig. 5a). This is due to the fact that temperature controls the latent heat released in the atmosphere because of the Clausius–Clapeyron effect (Held and Soden, 2006) and the meridional temperature gradient controls baroclinicity of the atmosphere (Stone, 1978). Additionally, another modulating factor is the vertical stratification of the atmosphere, as conditions of low stratification in the midlatitudes support stronger baroclinic activity for a given meridional temperature gradient (Holton, 2004). In the bistable region of the W state, MET has a flat response to increasing S^* and $[CO_2]$ and is to a large extent independent of T_s . This is because of the two contrasting and compensating effects: With increased T_s , water vapour concentration of the atmosphere increases, thus leading to the strengthening of the poleward latent heat fluxes. In addition, the increased T_s causes sea and continental ice as well as seasonal snow cover to retreat towards the poles, thus lowering the surface albedo gradient. This contributes negatively to the changes in MET , because it leads to a decrease in the baroclinicity. This extends the results obtained by Caballero and Langen (2005).

In the W regime, the boundary between the bistable and monostable regions approximately marks the point at which the Earth surface loses its permanent sea-ice cover, thus supporting the idea that the presence of bistability is intimately linked with the powerful ice-albedo feedback. For T_s larger than approximately 300 K, the meridional temperature gradient decreases at a far slower rate with increasing T_s . This essentially means that in this region the MET is controlled only by the availability of water vapour in the atmosphere. Therefore at temperatures above 300 K, MET becomes decoupled from the surface meridional temperature difference. This analysis consequently shows how important the involvement of the hydrological cycle is in the magnitude of the meridional heat transport and, moreover, it indicates that when going from warm to very warm climates the hydrological cycle becomes the dominant climatic feature, leading to strong positive dependence of the meridional heat transport on the surface temperature. Our results agree with the findings of Caballero and Langen (2005), who found in aqua planet simulations, that there is little scope for

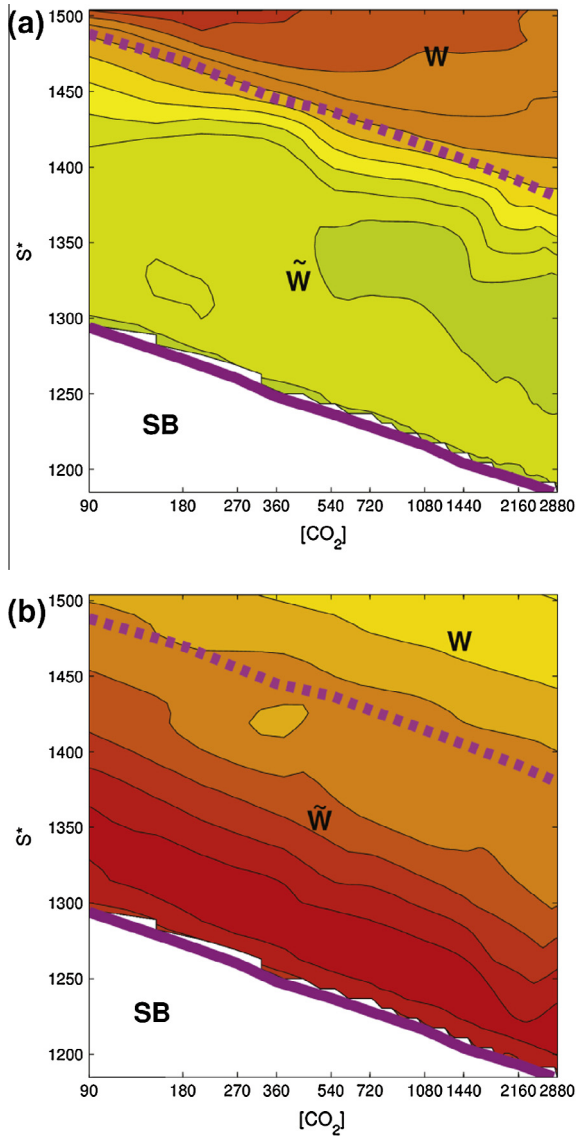


Fig. 5. Contour plot of: (a) meridional energy transport MET (PW, $1 PW = 10^{15} W$) and (b) average rate of dissipation of kinetic energy $\tilde{W}$ ($W m^{-2}$) as a function of S^* and the $[CO_2]$ for the W states.

reducing the meridional temperature gradient further once the sea-ice and snow have melted. For a constant meridional temperature gradient, an increase in the meridional latent heat fluxes with increasing global mean temperature was shown.

The conclusions drawn above find further support when looking at the mid-latitudes vertical temperature difference, ΔT_v , defined as the mean temperature difference between the surface and the 500 h Pa level (see Fig. 6b) and which provides a rough measure of the stratification of the atmosphere in the baroclinically active region. The vertical temperature difference is largest along a band of the CS space centred half way through the bistable region in the direction of S^* . Our current climate conditions would appear to be positioned at the centre of the band peak where the vertical temperature difference is maximised, implying conditions of reduced vertical stratification. For climates colder than present, increasing surface temperature causes the melting of sea-ice and seasonal snow cover, so that the ensuing decrease in surface albedo (leading to increased surface absorption) accounts for the increasing vertical temperature difference. Instead, in climates colder than present, the decrease in equatorial vertical temperature difference

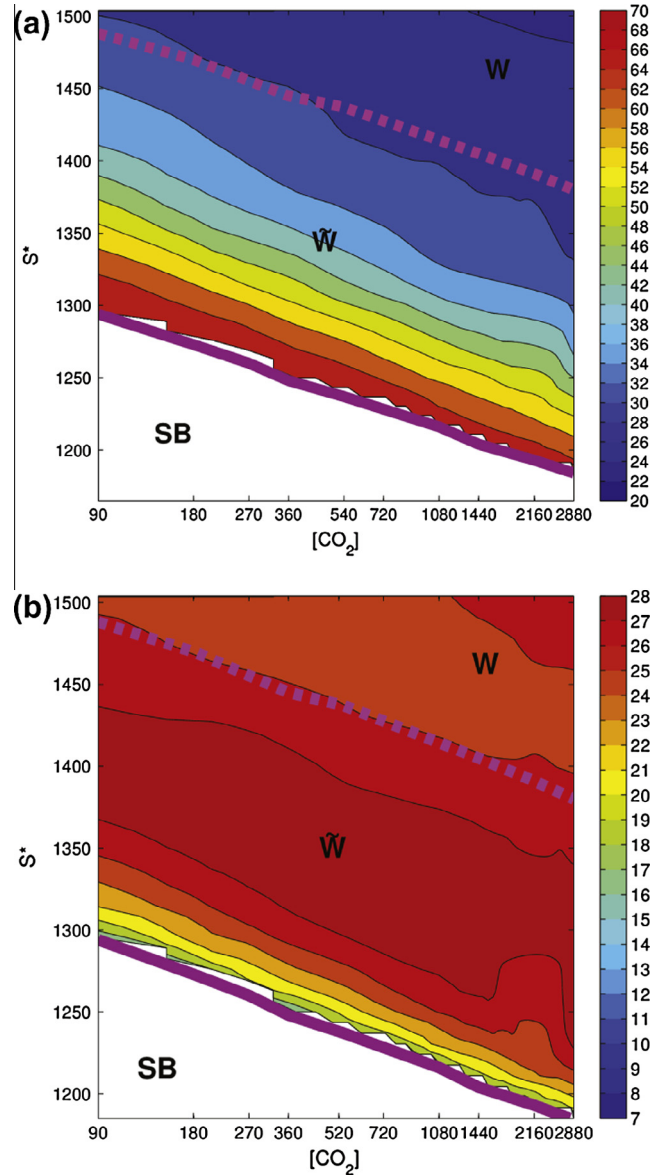


Fig. 6. Contour plot of: (a) meridional temperature gradient (K) and (b) midlatitude vertical temperature difference (K) as a function of S^* and the $[CO_2]$ for the W states.

with T_s can be understood in terms of increased moist convection from warmer surface temperatures, resulting in an increase of moisture fluxes to the upper atmosphere, which then condense and release latent heat.

In the bistable region, for the reasons discussed above, we expect to find a pronounced weakening of the dynamics of the climate system with increasing surface temperature. We test this hypothesis by computing the strength of the Lorenz energy cycle (Fig. 5b), which is equal to the average rate of dissipation of kinetic energy, and the Carnot efficiency of the system (Fig. 7a), which measures, instead, how far from equilibrium the system is. The dissipation, similarly to the meridional temperature gradient, decreases monotonically with the surface temperature, and reaches its largest value just at the cold boundary of the W manifold. The efficiency is maximised along a band of climate states characterised by slightly higher temperatures. This region in the CS space lies about half way between the peak in the meridional and vertical temperature gradients, since a large value of efficiency, by its very definition, requires a compromise between maximising these two gradients. The efficiency decreases monotonically with increasing

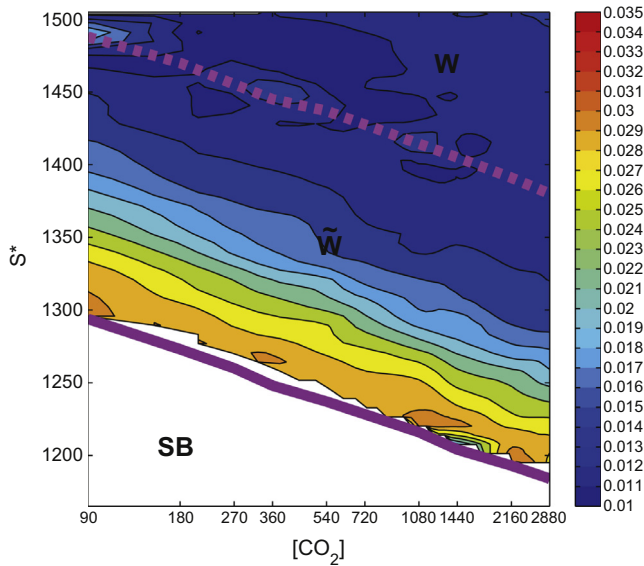


Fig. 7. Contour plot of the efficiency as a function of S^* and the CO_2 concentration for the W states.

$[CO_2]$ or S^* : warmer climates are characterised by smaller temperature differences, since the transport of water vapour acts as a very efficient means for homogenising the temperature across the system. The system has reduced ability to produce mechanical work and is characterised by very strong irreversible processes, as illustrated by the very large values of $\tilde{S}_{mat}$ realised in these conditions. Consequently the value of the parameter of irreversibility α increases as conditions become warmer and warmer (not shown); see also Lucarini et al. (2010b).

4.2. The Snowball state

The SB state is intrinsically simpler than the W state because the hydrological cycle has a negligible influence. This is due to the fact that atmospheric temperatures are so low that we have in all cases an almost dry atmosphere. As a result we have, e.g. that, as opposed to the W state, the parameter α is small (~ 1) and weakly dependent on the value of the parameter S^* and $[CO_2]$ (not shown); see also Lucarini et al. (2010b).

Moreover, in the SB state the meridional gradients (and not only the globally averaged values) of albedo are very low and depend weakly on S^* and the $[CO_2]$, because under all conditions the sea surface is frozen almost everywhere and the continents are covered by ice and snow. The meridional heat transport (Fig. 8a) is much smaller than for the corresponding W states, because in the SB state a large fraction of the radiation is reflected back to space and the meridional albedo gradients are small, so that the energy imbalance between low and high latitudes is small. Throughout the SB state, increases in S^* and $[CO_2]$, lead to an increase in T_s and $\tilde{S}_{mat}$. This is accompanied by increases in the meridional heat transport and in the dissipation of kinetic energy (Fig. 8b), which implies that with increased meridional heat transport the intensity of the circulation also increases. However we find that the meridional temperature difference (not shown) has a very weak dependence on changing $[CO_2]$ and S^* , as it varies by only 4 K across the explored parameter range. Therefore, the changes in the baroclinicity of the system which lead to changes in the meridional heat transport and the intensity of the Lorenz energy cycle, cannot be due to changes in the meridional temperature difference. The lack of large variations in the meridional surface temperature profile are essentially due to the fact that the net input of shortwave radiation is kept almost fixed by the constant

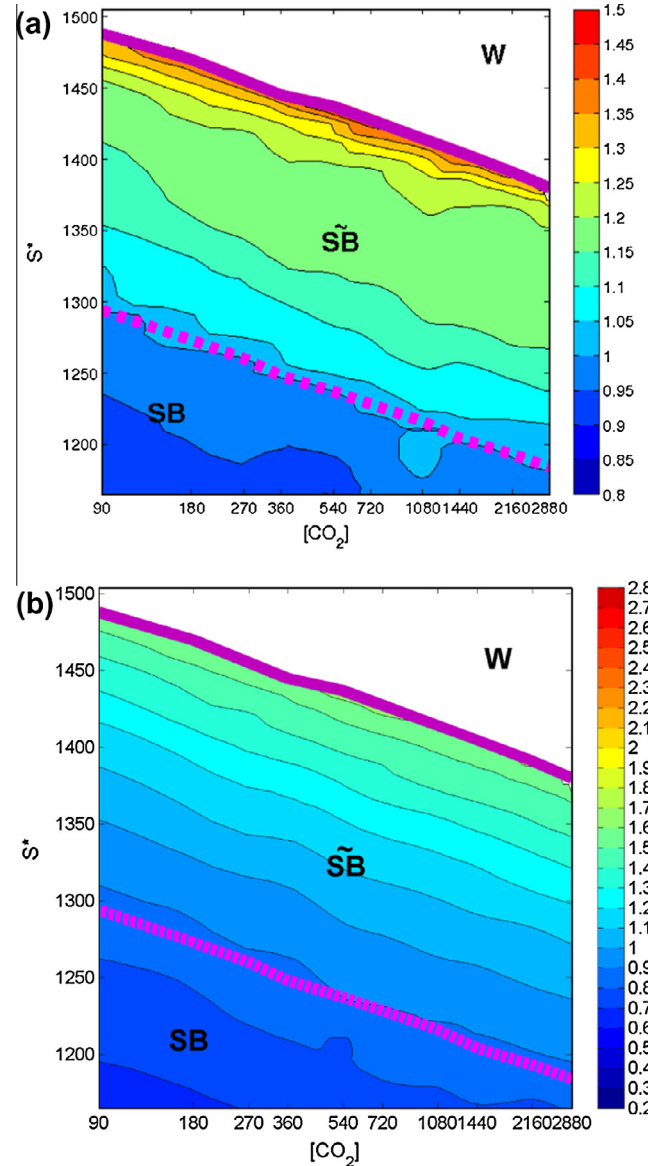


Fig. 8. Contour plot of: (a) meridional energy transport MET (PW) and (b) average rate of dissipation of kinetic energy ($W m^{-2}$) as a function of S^* and the $[CO_2]$ for the SB states.

surface albedo and by the virtue of almost no cloud cover. The system therefore can be seen as rather rigid, as changes in the absorbed radiation are almost exactly compensated for by changes in the meridional heat transport. The re-equilibration mechanism must then contain elements which are not present in the classic baroclinic adjustment (Stone, 1978). We find that the vertical temperature gradient in the midlatitudes (Fig. 9b), increases substantially with increased value of S^* and $[CO_2]$, as an effect of the increased absorption of radiation near the surface. The reduced vertical stratification leads to more pronounced baroclinic activity even for a fixed meridional temperature gradient, thus leading to an increase in MET and in the intensity of the Lorenz energy cycle. The argument is in this case quite straightforward because we are considering an almost dry atmosphere. The efficiency has a dependence on S^* and $[CO_2]$ which, as in the W case, is intermediate between those of the meridional and vertical temperature gradient fields. In the case of the SB state, the main signature is given by the vertical gradient (Fig. 9b). This further reinforces the idea that the investigation of meridional temperature gradients is not en-

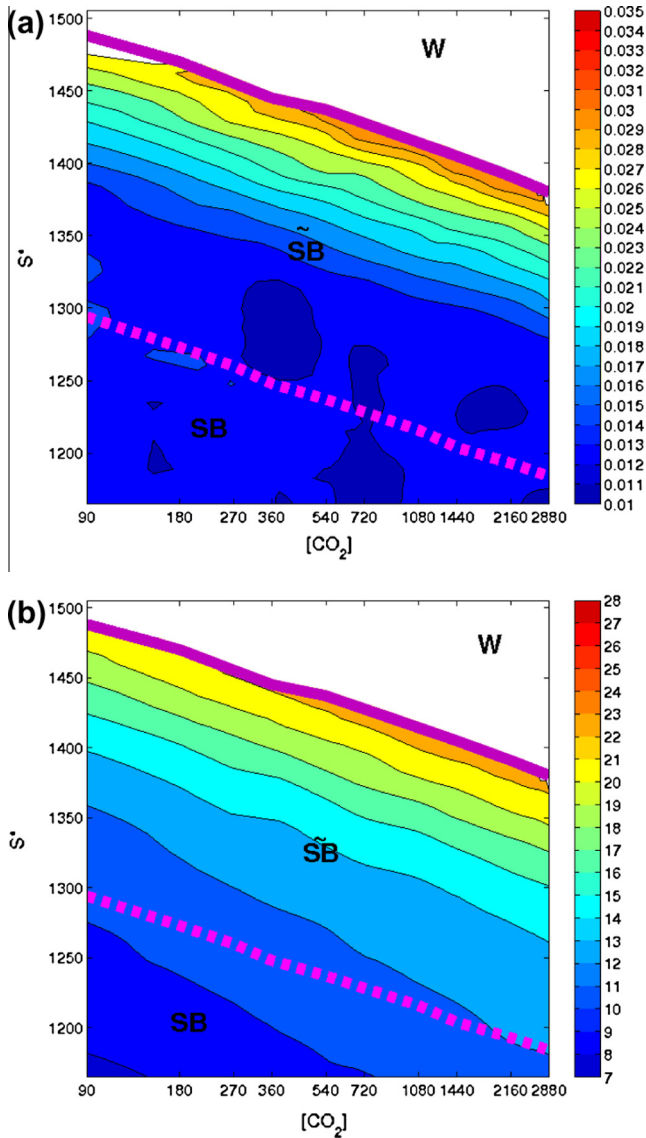


Fig. 9. Contour plot of (a) the Carnot efficiency η and (b) midlatitude vertical temperature difference (K) as a function of S^* and the $[CO_2]$ for the SB.

ough to grasp the mechanisms through which the system generates available potential energy and material entropy production (Lucarini et al., 2011).

4.3. Transition and comparison between manifolds

Up to now, the W and SB states have been characterised as two entirely distinct climate regimes, and have underlined that the basic mechanisms of re-equilibration are rather different. In this subsection we would like to present some ideas aimed at making sense of the transitions between the two states occurring when we get close to the boundaries of the upper and lower manifolds. As has been seen, the $W \rightarrow SB$ transition is associated with a large decrease in T_s , $\bar{S}_{mat}$ and MET . This is intimately related to the fact that whereas in the W state the hydrological cycle plays a major role in the climate dynamics, in the SB state the hydrological cycle is almost absent. Nonetheless, this does not say much about the processes leading to the transition between the two states, or better, describing how one of the attractors disappears.

We have also discovered that the usual dynamical indicators of the atmospheric state, i.e. the meridional temperature gradient and

the vertical stratification do not necessarily indicate whether or not we are close to an irreversible transition of the system, e.g. by signaling something equivalent to a loss of elasticity of the system. In this regard, it is much more informative to observe how the efficiency behaves near the transitions. We find that, as a general rule, each transition is associated to a notable decrease (more than 30%) of the efficiency of the system, and the closer the system gets to the transition in the CS space, the larger is the value of the efficiency. This can be interpreted as follows. If the system approaches a bifurcation point, its positive feedbacks become relatively strong and the negative feedbacks, which act as re-equilibrating mechanisms, become less efficient. As a result, the differential heating driving the climate is damped less effectively, and the system is further from equilibrium, since larger temperature differences are present. Therefore, the system produces more work, thus featuring an enhanced Lorenz energy cycle and a stronger circulation. At the bifurcation point, the positive feedbacks prevail and the circulation, even if rather strong, is not able to cope with the destabilising processes, and the transition to the other manifold is realised. The new state is, by definition, more stable, and thus closer to equilibrium. The decreased value of the efficiency is exactly the marker of this property. This confirms what has been proposed by Lucarini et al. (2010b) but in much greater generality and can be conjectured to be a rather general property of non-equilibrium systems featuring structural instabilities.

5. Parameterisations

In the previous sections, we have presented a systematic investigation of the thermodynamical properties of the planet in the SB and W states, obtained by exploring the CS parametric space. As mentioned above, in many cases the main quantity controlling the thermodynamic properties is the average surface temperature T_s only. This statement can be substantiated by looking at Figs. 3–9 and observing that in most cases the isolines of the depicted thermodynamic quantities are approximately parallel, both in the W and SB state, to the isolines of T_s . This suggests the possibility of establishing approximate empirical laws of the form $\Gamma_{SB}(S^*, [CO_2]) \approx \Gamma_{SB}(T_s(S^*, [CO_2]))$ and $\Gamma_W(S^*, [CO_2]) \approx \Gamma_W(T_s(S^*, [CO_2]))$, where Γ is a thermodynamical property such as, e.g. entropy production, and the lower index refers to whether we are in the SB or W state. For a given Γ , the empirical laws will be in general

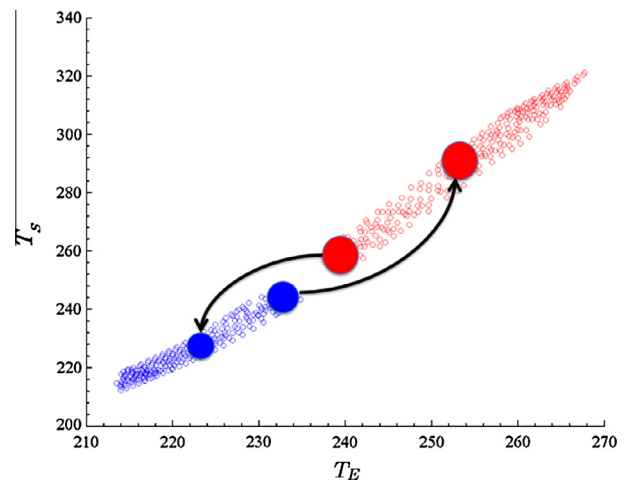


Fig. 10. Surface temperature T_s (K) vs. emission temperature T_E (K) for the W and SB states (each circle represents a simulation). We have marked the transitions $W \rightarrow SB$ (red dot–arrow–blue dot) and $SB \rightarrow W$ (blue dot–arrow–red dot). Same convention is used in Figs. 11–14. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

different in the SB vs. the W state. This result is quite interesting in a classical perspective of climate dynamics, where it is customary to parameterise large scale climate properties as a function of the surface temperature, especially when constructing simple yet meaningful models (Saltzman, 2002).

We want to complement this approach by testing whether it is possible to construct empirical laws expressing the thermodynamical properties of the system as a function of its emission temperature $T_E = (LW_{TOA}/\sigma)^{1/4}$ – where LW_{TOA} is the area- and time-averaged outgoing long wave radiation at the top of the atmosphere and σ the Stefan–Boltzmann constant, which is a quantity of more readily astrophysical interest and more fundamental physical significance. Therefore, we would like to be able to observe well-defined functions of the form $\Gamma_{SB}(S^*, [\text{CO}_2]) \approx \Gamma_{SB}(T_E(S^*, [-\text{CO}_2]))$ and $\Gamma_W(S^*, [\text{CO}_2]) \approx \Gamma_W(T_E(S^*, [\text{CO}_2]))$, which, given the observation made above, requires also the existence of an approximate relationship $T_E \approx T_E(T_S)$.

Indeed, Fig. 10 shows that it is possible to observe an empirical, monotonic relationship between T_S and T_E in a vast range of climate states, including both the W and SB states. Obviously, the correspondence between the two variables is not perfect, so that for each value of T_S it is possible to have climates featuring a range of values of T_E , and vice versa. The spread associated with the ensemble points is of order 10 K and 5 K for T_E , and 20 K and 10 K for T_S , in the W and SB states respectively. A range of values

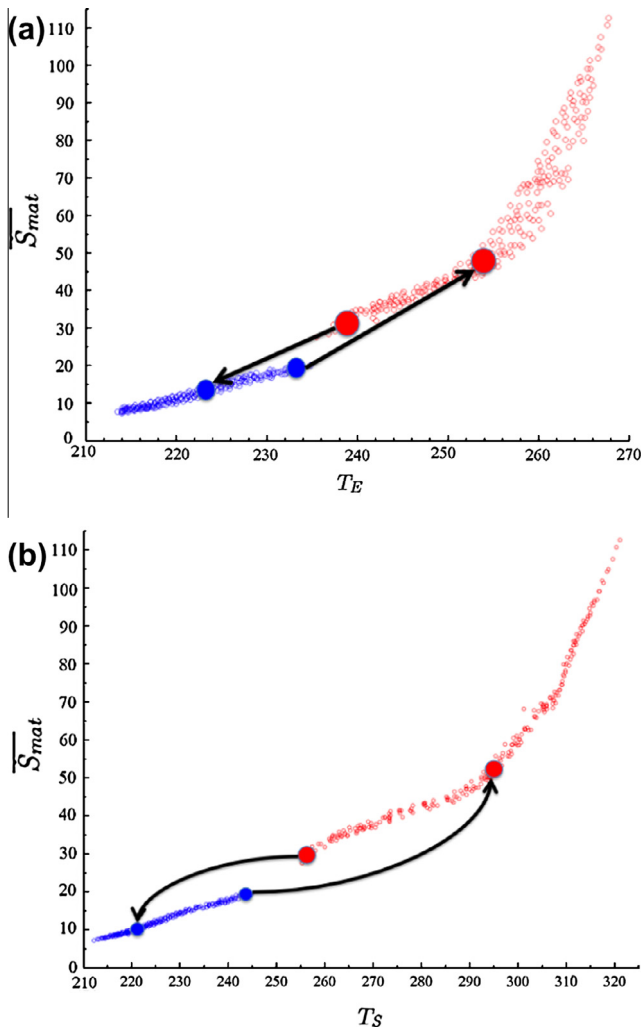


Fig. 11. Material entropy production $\bar{S}_{mat}$ ($10^{-3} \text{ W m}^{-2} \text{ K}^{-1}$) vs. (a) emission temperature T_E (K) and (b) surface temperature T_S (K).

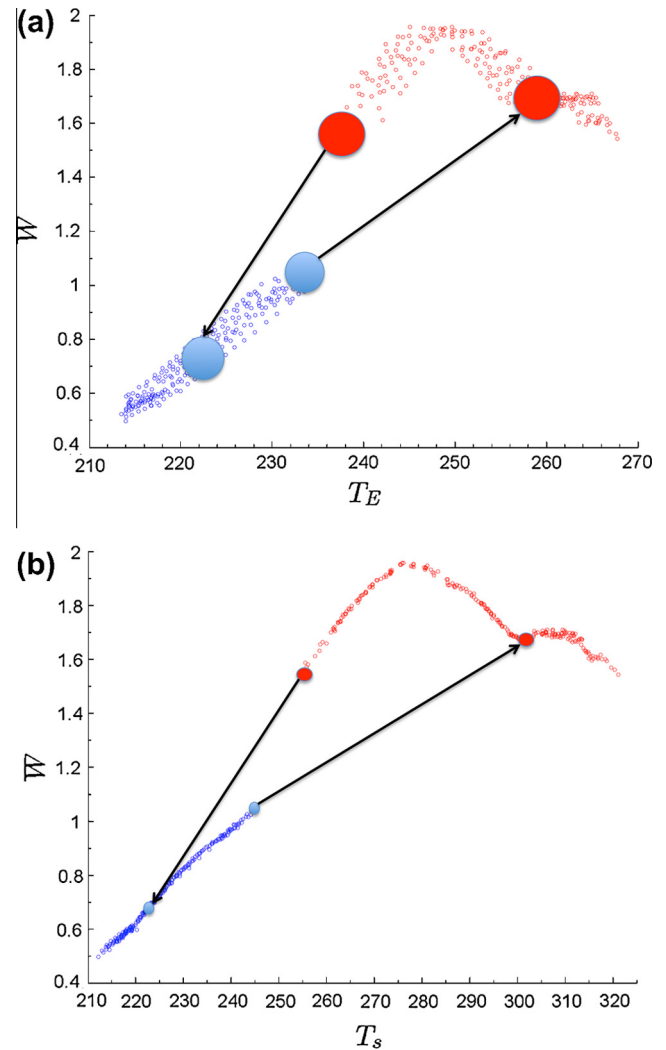


Fig. 12. Lorenz Energy cycle strength $\bar{W}$ (W m^{-2}) vs. (a) emission temperature T_E (K) and (b) surface temperature T_S (K).

of T_E (230–240 K) and T_S (240–260 K) are not permitted by the system as a result of the ice-albedo instability. We can now proceed to see to which extent the quantities T_S and T_E can be used as a good predictor of the thermodynamic properties of the planet and compare their skill with T_S . The following description is, on purpose, purely qualitative, because we mostly want to show to what extent T_E and T_S can be used as predictors, rather than discussing in detail what are the appropriate transfer functions.

Fig. 11a and b shows that a clear monotonic relationship exists between T_E , T_S and the material entropy production $\bar{S}_{mat}$, with higher values of T_E and T_S associated with larger values of $\bar{S}_{mat}$. It is important to note that while the values of $\bar{S}_{mat}$ and T_S are related to a high degree of accuracy, the functional dependence between T_E and $\bar{S}_{mat}$ is less well defined (the spread of points is much wider than in the T_S case, and especially so in very warm states). Note also that the dependences, while monotonic in both cases, are very different between the W and SB states. The consensus is that the greater the energy absorbed by the planet (larger values of T_E), the more intense is the production of material entropy. Note that this is indeed not the case when considering the intensity of the Lorenz energy cycle (Fig. 12a and b): higher absorption of energy corresponds to a weaker energy cycle in the W state and to a stronger cycle in the SB state. Therefore, the increase in the entropy production with T_E is due, in the SB case, to the increased dissipation

of kinetic energy, while in the W state it is due to the greatly enhanced hydrological cycle. In agreement with what has been discussed in the previous section, one can explain the contrasting responses of the W and SB states to increases in the absorption of energy by analysing the Carnot efficiency of the planetary fluid (Fig. 13). It is interesting to note that, at the transitions, the values of the efficiency are almost equal: $\eta_{W \rightarrow SB} \approx \eta_{SB, W} \approx 0.03$. Moreover, we observe that η saturates in the very cold regime of SB states and very warm regime of W (see Fig. 13) at $\eta_{sat, W} \approx \eta_{sat, SB} \approx 0.012$. Also note, that as with $\bar{S}_{mat}$, T_s does a better job than T_E , as a predictor of the other thermodynamic quantities considered here i.e. the Lorenz energy cycle and efficiency, since the spread of points is much narrower.

Now, the link between T_E , T_s and MET (Fig. 14) is discussed. We observe that in the SB regime there is a weakly positive relationship between T_E (T_s) and the transport, for the reasons described in the previous section. Instead, for a vast range of values of T_E (T_s) in the W regime, the transport is almost insensitive to T_E (T_s), as discussed thoroughly in Caballero and Langen (2005) and Donohoe and Battisti (2011, 2012). The rigidity of the climate system has been attributed by Donohoe and Battisti (2011, 2012) to the fact that the MET is mainly determined by planetary albedo and thus atmospheric composition rather than surface albedo and sea-ice coverage. However, such a mechanism ceases to exist when the ice albedo feedback becomes ineffective because of the disap-

pearance of sea-ice above a threshold value of $T_E \approx 255$ K ($T_s \approx 300$ K). Above this value, we observe a steep monotonic increase of the transport with temperature, because changes in the latent heat transport are mainly responsible for this behaviour. This agrees with the idea that the dynamics and sensitivity of a warm planet is, in some sense, dominated by the hydrological cycle. In this latter case, the skill of the two temperature quantities in parameterising the MET is comparable. Indeed, the details of the transfer functions will depend on various specific properties of the planetary system under investigation. The focus here is on the fact that our results suggest that it is possible to define such empirical relations. Finally, Fig. 15 describes how the sea-ice fraction can be parameterised as a function of the surface and emission temperature. As we see, the surface temperature is an almost exact predictor of the sea-ice fraction, whereas the emission temperature provides a noisier signal, especially in the W states. It is interesting to note that transition from the W to SB states occurs at a critical value of the sea-ice fraction of about 0.5, in broad agreement with previous results by Voigt et al. (2011). The freezing of the planet leads to the sea-ice fraction tending to unity, whereas the reverse SB $\rightarrow$ W transition has an even more dramatic change, because it goes from a completely frozen to a virtually ice free ocean. This elucidates even more so, the strength of the ice-albedo feedback.

Figs. 11–15 clearly define the transfer functions allowing to compute the thermodynamic properties of the planet from the va-

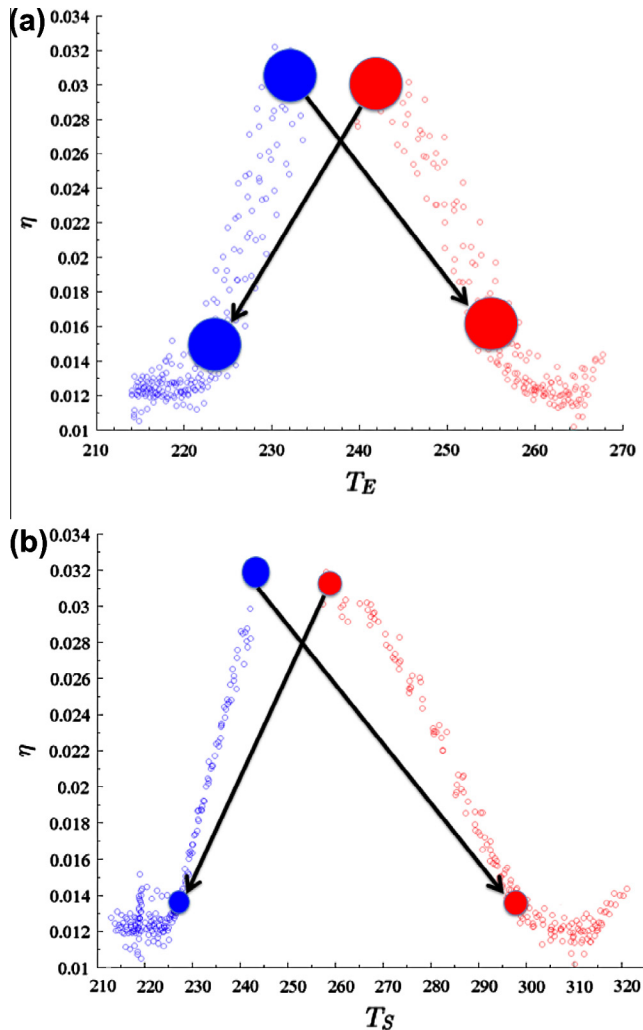


Fig. 13. Carnot efficiency η vs. (a) emission temperature T_E (K) and (b) surface temperature T_s (K).

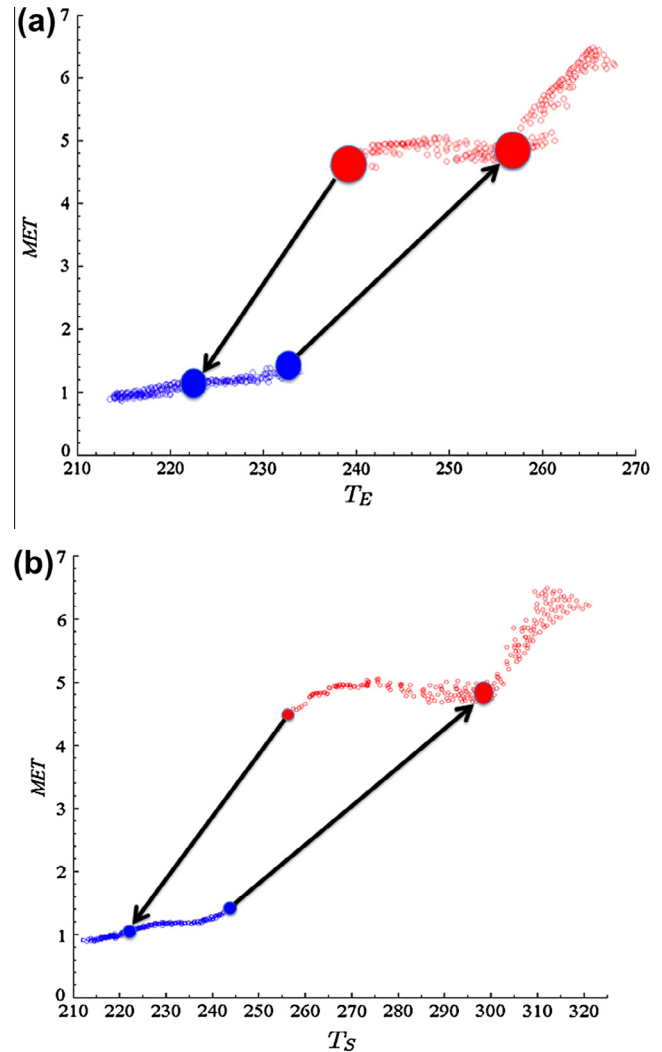


Fig. 14. Meridional heat transport index MET (in PW) vs. (a) emission temperature T_E (K) and (b) surface temperature T_s (K).

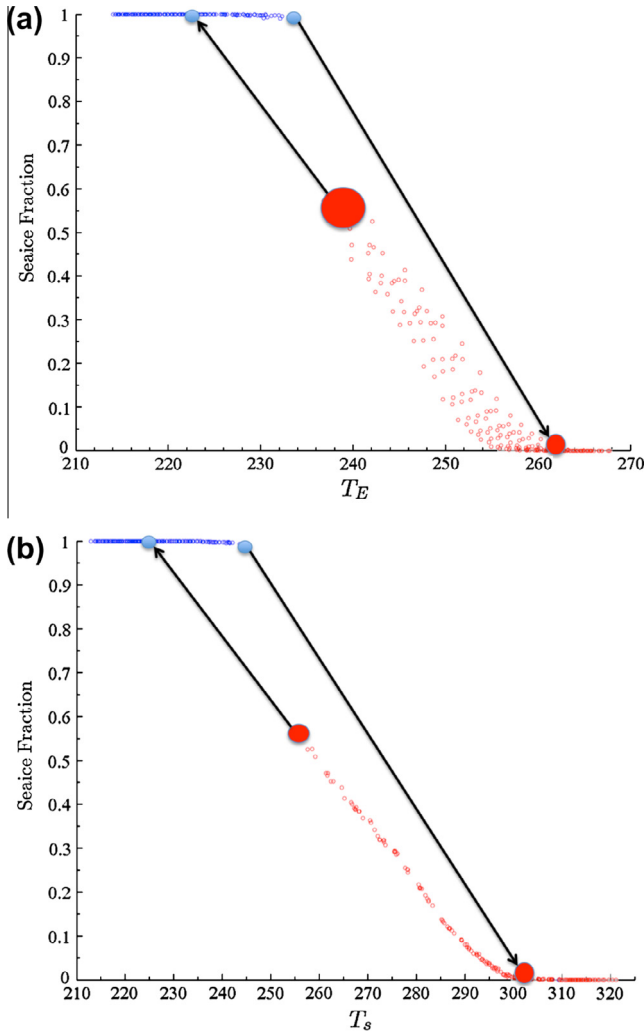


Fig. 15. Sea ice fraction (in %) vs. (a) emission temperature T_E (K) and (b) surface temperature T_s (K).

lue of T_E or T_s only. This is definitely intriguing because it paves the way for computing non-equilibrium properties from a quantity describing the zero-order radiation balance of the system at the top of the atmosphere and of the radiative–convective equilibrium at the surface (in the case of T_s). Figs. 11–15 leave various open questions which definitely need to be accurately addressed in future studies: (i) Why is it possible to use to a good degree of approximation the quantity T_E to parameterise the non-equilibrium thermodynamical properties of a system? (ii) Why does the surface temperature do an even better job (actually, astounding) in controlling such complex properties? (iii) Is the result where the efficiency is found to decrease at the bifurcation points for both $W \rightarrow SB$ and $SB \rightarrow W$, a transition of more general relevance?

6. Thermodynamics of a slowly rotating planet

The results presented in the previous sections are indeed not universal because they have been obtained making specific choices on some crucial ingredients such as the position and size of the continents, the radius of the planet, the eccentricity of its orbit, the orbital tilt and its rotation rate, among others. Obviously, it is virtually impossible to explore such a vast multidimensional parametric space, and one must come to terms with the fact that only partial explorations are possible (and future investigations will

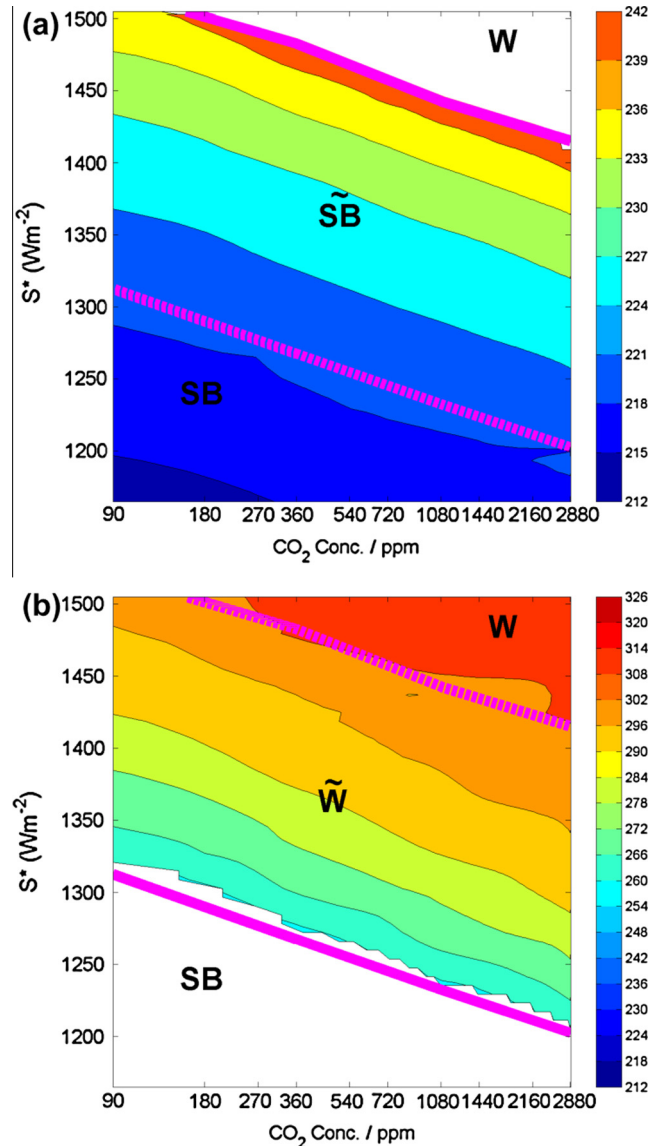


Fig. 16. Contour plot of $\overline{T_s}$ (K) as a function of S^* and the $[CO_2]$ for the lower SB (a) and upper W (b) manifolds for the slowly rotating planet.

deal with different axes of such parametric space). We make just a small step in this direction by testing the sensitivity of our results with respect to changes in the planetary rotation rate Ω . In particular, we investigate here the case where Ω is reduced by a factor of 2. As well known from dynamical meteorology, reducing the rotation rate *ceteris paribus* implies that the mid-latitude flow is more weakly constrained to being in the vicinity of geostrophic equilibrium, because the Rossby number is increased. Baroclinic instability is weakened because the Rossby deformation radius is increased, and, the meridional extent of the Hadley cell increases because angular momentum conservation constraints become less stringent. This causes a dramatic readjustment between the key mechanisms of poleward atmospheric transport and of the transformation of available into kinetic energy. Therefore, we have repeated some of the experiments described before with a lower value of the rotation rate. In particular, we have considered for $[CO_2]$ values of 90, 360, 1080 and 2880, while S^* has been modulated between 1510 W m^{-2} and 1150 W m^{-2} in steps of 30 W m^{-2} .

First, we note that also for such a lower value of Ω the system features bi-stability, and the overall structure of the phase space is only modestly altered. The $W \rightarrow SB$ and $SB \rightarrow W$ boundaries de-

scribed above are still approximately parallel straight lines in the (S^* , $\log[\text{CO}_2]$) plane, but both shifted to the right with respect to what was found previously. More specifically, $\text{SB} \rightarrow \text{W}$ and $\text{W} \rightarrow \text{SB}$ transition lines are shifted to higher values of S^* by approximately 30 W m^{-2} and 15 W m^{-2} respectively. Interestingly, the $\text{SB} \rightarrow \text{W}$ and $\text{W} \rightarrow \text{SB}$ transition lines are also in this case closely aligned along isolines of the surface temperature, and with approximately the same value observed for the regular rotation rate. This is reported in Fig. 16.

We now explore to what extent we can replicate the results shown in Figs. 11–15. The first notable observation is that for a slowly rotating planet, non-equilibrium thermodynamical variables like η , S_{mat} and MET can still be expressed to a high degree of accuracy as functions of T_S or T_E . The second aspect is that the empirical laws we find in this case are analogous to what was found for the terrestrial value of Ω .

In particular, we find that the empirical parametrization of MET is virtually identical for both cases (Fig. 17), even if corresponding climate states for different values of Ω are very different. In general we observe that in the case of weaker rotation, the meridional streamfunction extends further polewards and similarly the jet

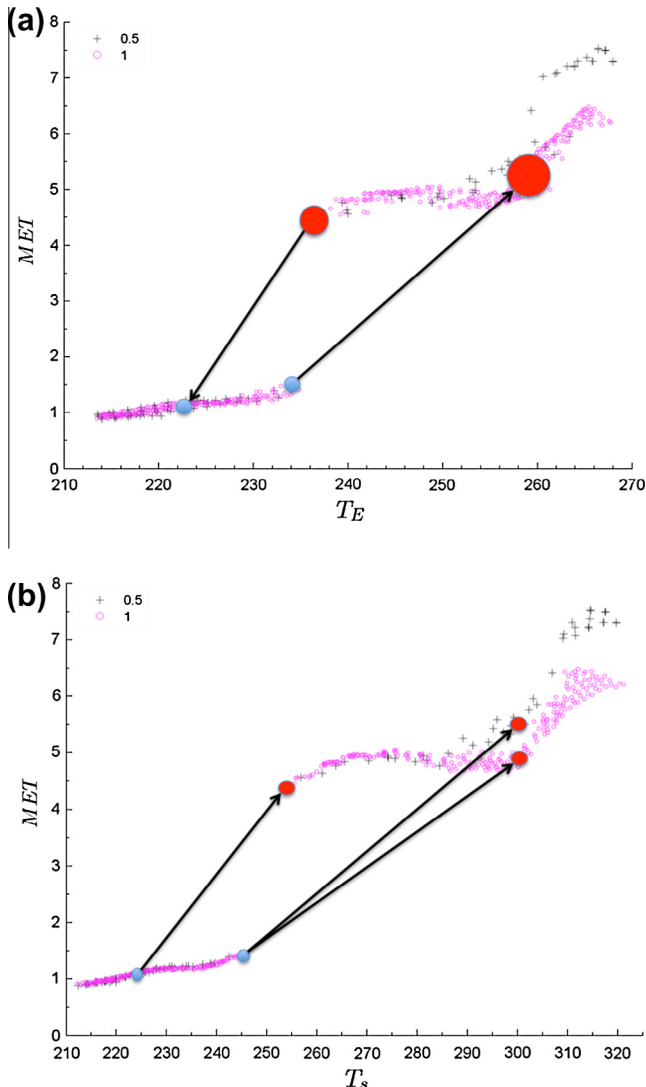


Fig. 17. Meridional heat transport index MET (in PW) vs. (a) emission temperature T_E (K) and (b) surface temperature T_S (K) for $\Omega/\Omega_E = 1$ (magenta) and $\Omega/\Omega_E = 0.5$ (black). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

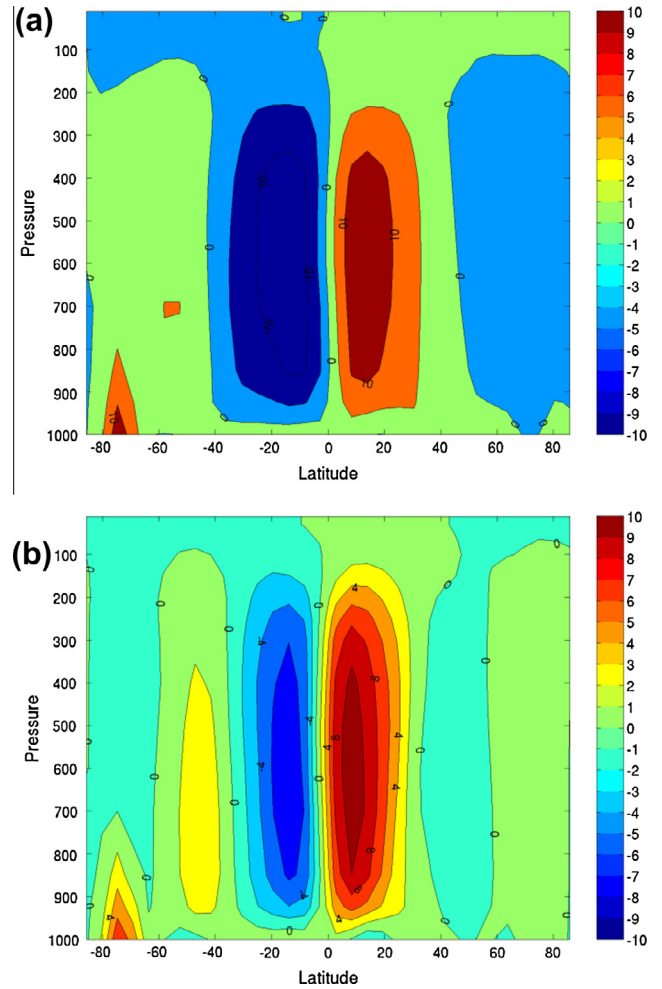


Fig. 18. Zonal mean streamfunction profile for: (a) $\Omega/\Omega_E = 0.5$ and (b) $\Omega/\Omega_E = 1$. Solar constant and CO_2 concentration set to present terrestrial level.

stream is shifted polewards, as expected (see Figs. 18a, b, 19a and b). The system with faster rotation features stronger eddy heat transports, as its mid-latitude dynamics are dominated more so by baroclinic conversion. Despite these relevant differences, within this range of rotation rates the total MET is not highly sensitive to the specific mechanisms of transport.

An even more fundamental indication of the robustness of the thermodynamic parameterisations proposed above is that the functional dependence of S_{mat} with respect to T_S is almost unchanged when considering the slower rotation case (Fig. 20), both in the SB and W regimes. Therefore, the global effect of the irreversible processes, is not highly sensitive to the specific dynamical features determining mixing and dissipation throughout the domain. Moreover, the other basic measure of the non-equilibrium properties of the system, i.e. the efficiency η , has a very weak dependence on the rotation rate, as shown in Fig. 21. Finally, the Lorenz energy cycle (Fig. 22) also behaves similarly in both cases, though the intensity of the more slowly rotating planet is biased low, as the atmospheric features have in this case larger spatial scales and therefore undergo weaker viscous dissipation. Finally, we observe that the sea-ice fraction is also weakly influenced by the slower rotation rate (Fig. 23).

These results are encouraging as they suggest that the empirically derived functional relations between thermodynamic quantities and averaged temperatures are reasonably robust with respect to reductions in the rotation rate. While we do not claim any strong

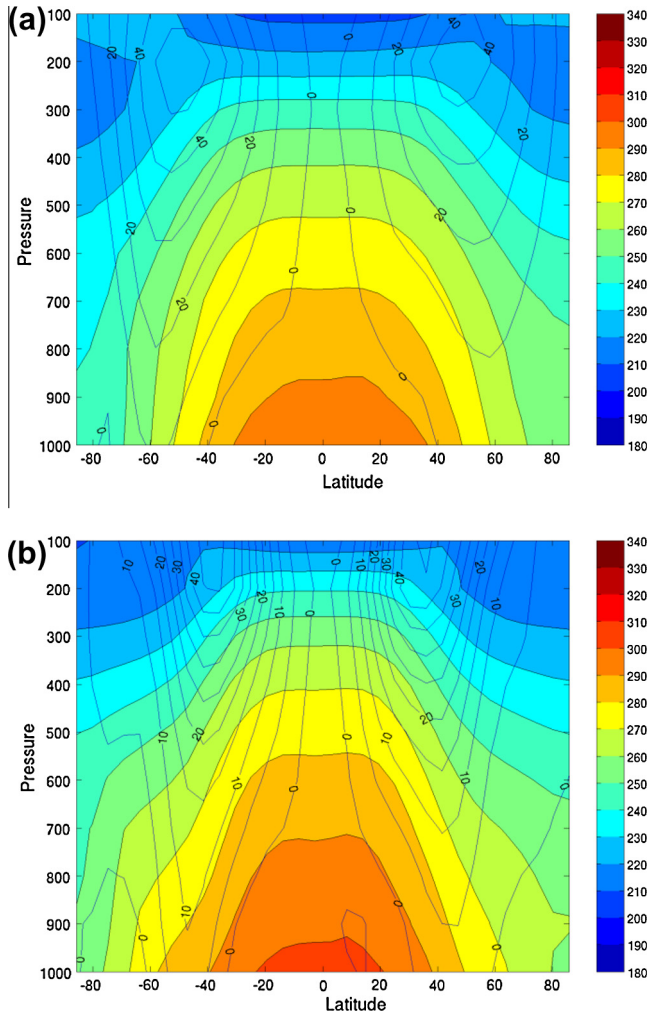


Fig. 19. Zonal mean distribution of westerly winds (ms^{-1}) and atmospheric temperature for: (a) $\Omega/\Omega_E = 0.5$ and (b) $\Omega/\Omega_E = 1$. Solar constant and CO_2 concentration set to present terrestrial level.

degree of universality in our results, such robustness hints at the fact that our proposed approach has some degree of generality.

7. Summary and conclusions

Motivated by the recent discoveries of extrasolar planets, we have re-examined for the case of an Earth-like planet, whether being in the habitable zone is a sufficient condition for ensuring the presence of liquid water at the surface. This has been done by considering the case of a planet with the same geography and topography of the present Earth and have performed an extensive parametric study by exploring the effect of changing the solar irradiance and the atmospheric opacity (the latter modulated by the CO_2 concentration).

Within a rather wide parametric space, which includes the present conditions, the climate is multistable, i.e. there are two coexisting attractors, one characterised by warm conditions, where the presence of sea-ice and seasonal snow cover is limited (W state), and the other by a completely frozen sea surface, the so-called Snowball (SB) state. Our results, obtained using the climate model PlaSim, confirm and extend what has been obtained in various studies performed with models of various degrees of complexity. We point the reader to [Pierrehumbert et al. \(2011\)](#) and [Lucarini et al. \(2010b\)](#) for an extensive discussion. In this regard, the main

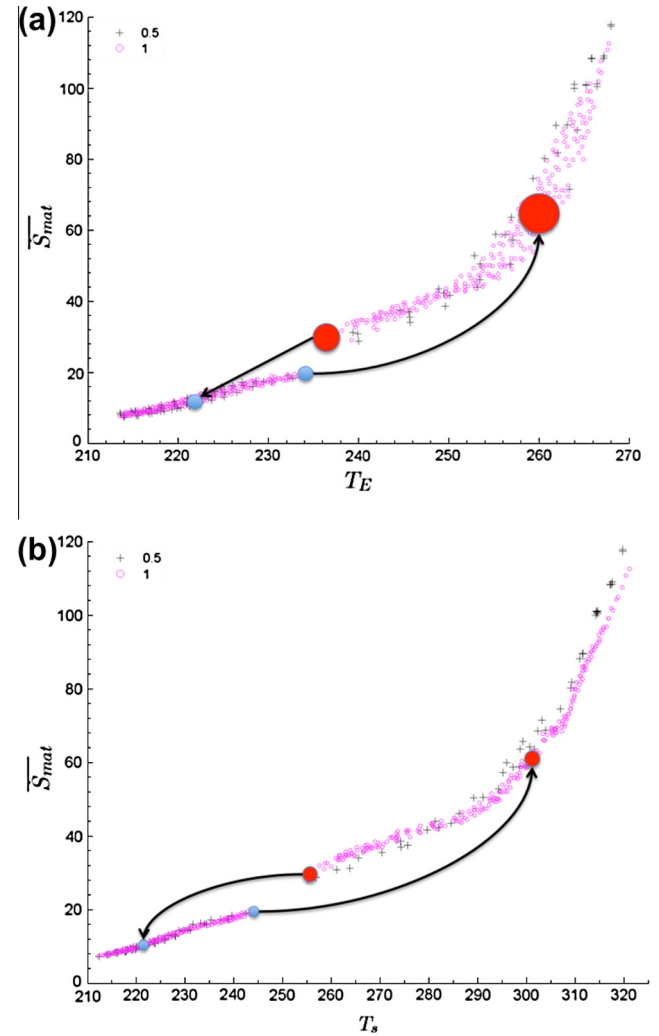


Fig. 20. Material entropy production $\overline{S_{mat}}$ ($10^{-3} \text{ W m}^{-2} \text{ K}^{-1}$) vs. (a) emission temperature T_E (K) and (b) surface temperature T_s (K) for $\Omega/\Omega_E = 1$ (magenta) and $\Omega/\Omega_E = 0.5$ (black). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

improvement of this work is that a two-dimensional parametric space is explored (whereas usually variations in the solar constant and atmospheric opacity are considered separately), which allows the gathering of more complete information on the possible states of the climate system and actual mechanisms relevant in a paleoclimatological and planetary context. Since our exploration of the parametric space is limited and we consider specific boundary conditions typical of our planet, our investigation is not fully comprehensive nor universal. Nonetheless, we maintain that our study provides methodological hints and qualitative results of wide interest.

For all considered values of $[\text{CO}_2]$, which range from 90 to 2880 ppm, the width of the bistable region is about 200 W m^{-2} in terms of the value of the solar constant, and its position depends linearly on the logarithm of the $[\text{CO}_2]$, being centred around smaller values of the solar constant for increasing opacity of the atmosphere, shifting by about 15 W m^{-2} per doubling of CO_2 concentration. The W state is characterised by surface temperature being 40–60 K higher than in the SB state. In terms of the material entropy production, it is larger by a factor of 4 (order of $40\text{--}60 \times 10^{-3} \text{ W m}^{-2} \text{ K}^{-1}$ vs. $10\text{--}15 \times 10^{-3} \text{ W m}^{-2} \text{ K}^{-1}$). The boundaries of the bistable region are approximately isolines of the globally averaged surface temperature, and in particular, the warm

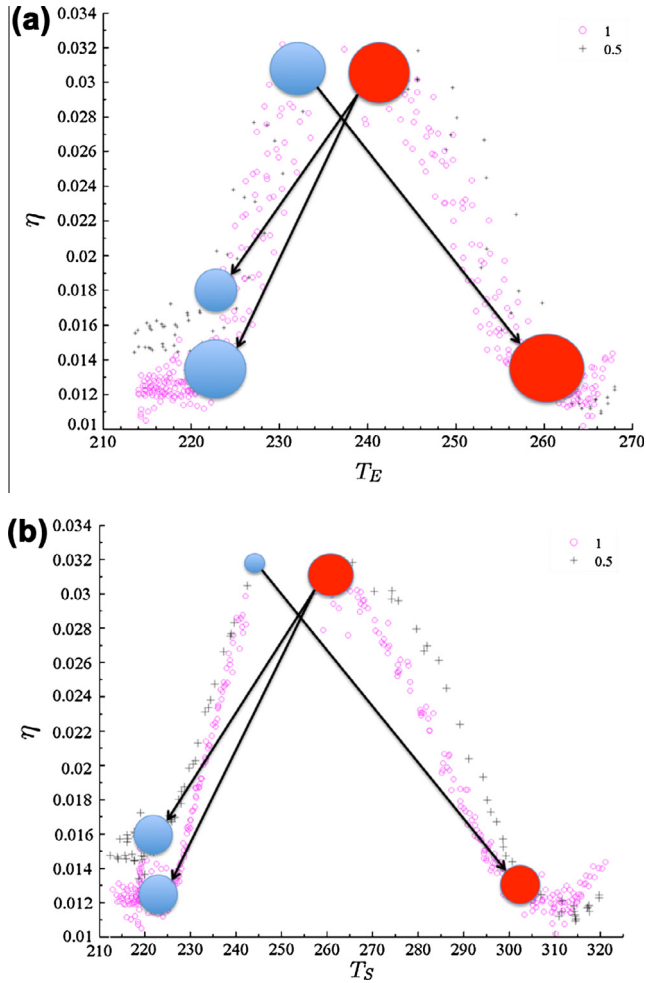


Fig. 21. Carnot efficiency vs. (a) emission temperature T_E (K) and (b) surface temperature T_S (K) for $\Omega/\Omega_E = 1$ (magenta) and $\Omega/\Omega_E = 0.5$ (black). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

boundary, beyond which the SB state cannot be realised, is characterised by the vanishing of the permanent sea-ice cover in the W regime. This reinforces the idea that the ice-albedo feedback is the dominant mechanism necessary for obtaining the multistability properties of the climate system.

The thermodynamical and dynamical properties of the two states are largely different, as if we were discussing two entirely different planets. In the W state the climate is dominated by the hydrological cycle and latent heat fluxes are prominent in terms of redistributing energy in the system and as the main contributor to the material entropy production. The SB state is an extremely dry climate, where heat transport is realised through sensible heat fluxes and entropy is mostly generated through the dissipation of the kinetic energy. The dryness of the SB atmospheric states also explain why the climate sensitivity is much smaller.

In the bistable region of the W states, the meridional heat transport is rather constant, as the contrasting effect of the increasing latent heat fluxes driven by increasing surface temperature, coupled with the reduction in the baroclinicity due to the decrease in the meridional temperature gradient compensate almost exactly. In the W state beyond the bistable region, the meridional heat flux increases quite dramatically with the surface temperature (Figs. 6a and 14) since the compensating albedo mechanisms are shut off as sea-ice is completely removed from the surface. In the SB states, increased incoming radiation or increased $[\text{CO}_2]$ leads

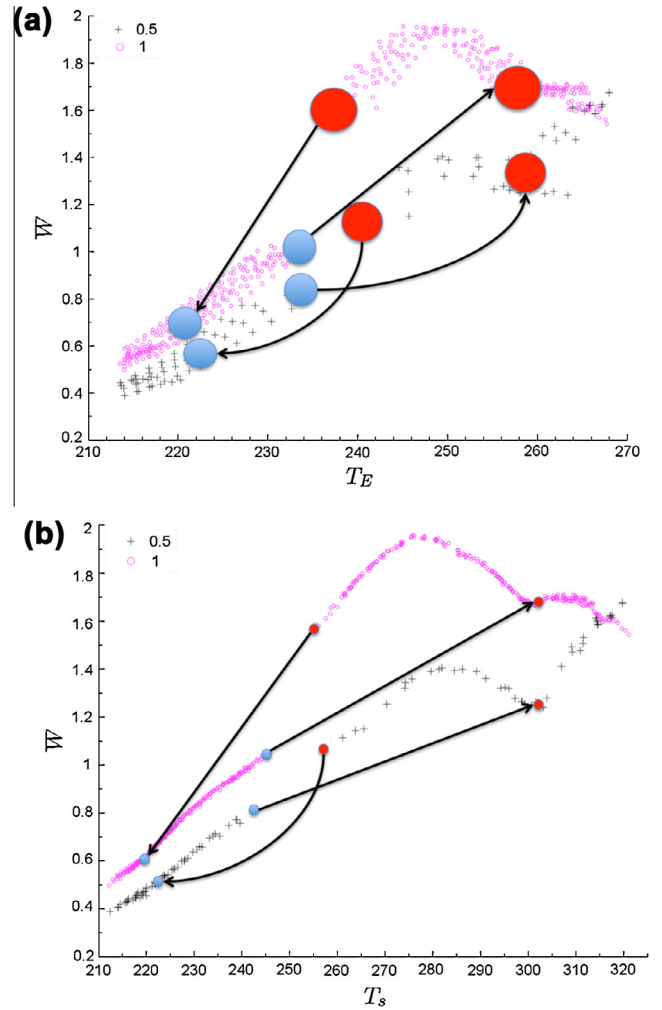


Fig. 22. Lorenz energy cycle strength $\bar{W}$ (W m⁻²) vs. (a) emission temperature T_E (K) and (b) surface temperature T_S (K) for $\Omega/\Omega_E = 1$ (magenta) and $\Omega/\Omega_E = 0.5$ (black). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

to increases in the meridional heat transport. In this case, the water vapour plays little part, and, somewhat surprisingly, the meridional temperature difference has also a rather flat response to the parametric modulation. In this case, the dominant mechanism determining the properties of the meridional heat transfer is the change in the vertical stratification, which becomes weaker for warmer climate conditions because the atmosphere is more strongly heated from below. This implies that the atmospheric circulation strengthens for increasing values of the solar constant and of $[\text{CO}_2]$. In fact, the Lorenz energy cycle becomes stronger for warmer climate conditions, and the Carnot-like efficiency of the climate system has an analogous behaviour. The opposite holds for the W state, where the intensity of the Lorenz energy cycle and the efficiency decreases for warmer conditions, the reason being that the water vapour becomes more and more efficient in homogenising the system and destroying its ability to generate available potential energy.

A general property which has been found is that, in both regimes, the efficiency increases when we get closer to the bifurcation point and at the bifurcation point the transition to the newly realised stationary state is accompanied by a large decrease in the efficiency. This can be framed in a rather general thermodynamical context: the efficiency gives a measure of how far from equilibrium the system is. The negative feedbacks tend to counter-

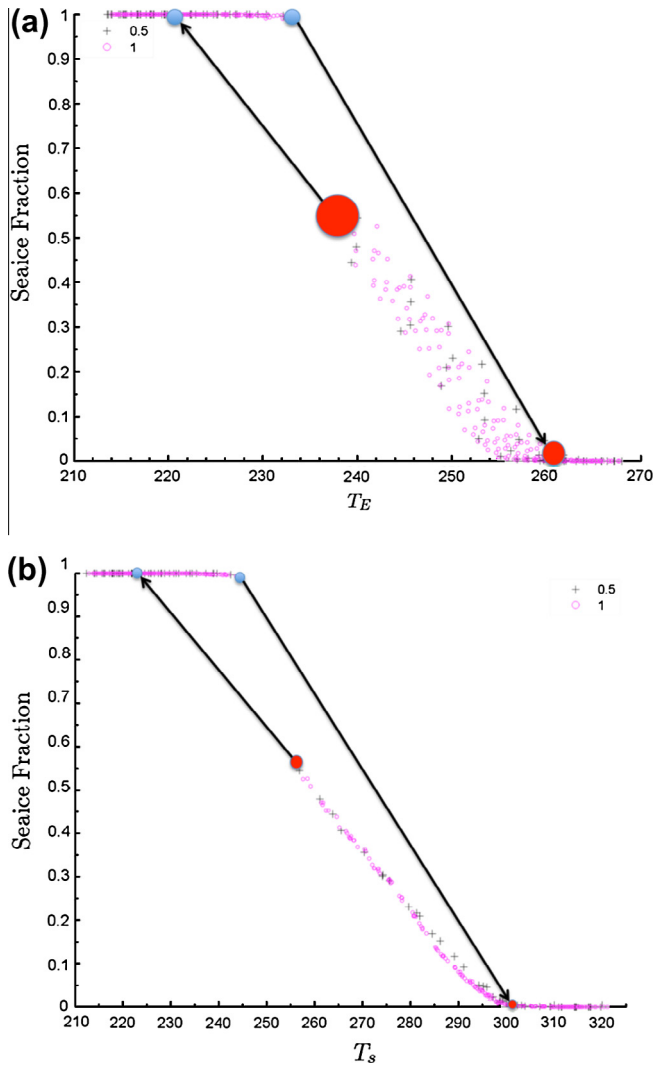


Fig. 23. Sea ice fraction (in %) vs. (a) emission temperature T_E (K) and (b) surface temperature T_s (K) for $\Omega/\Omega_E = 1$ (magenta) and $\Omega/\Omega_E = 0.5$ (black). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

act the differential heating due to the sun's insolation pattern, thus leading the system closer to equilibrium. At the bifurcation point, the negative feedbacks are overcome by the positive feedbacks, so that the system makes a global transition to a new state, where, in turn, the negative feedbacks are more efficient in stabilizing the system. On a more phenomenological note, the transition from the W to SB states occurs at a critical value of the sea-ice fraction of about 0.5. This agrees with previous findings (Voigt and Pierrehumbert, 2011). After the transitions, the sea-ice fraction becomes unitary. When considering the reverse tipping point, the transition is even more dramatic as the sea-ice fraction changes abruptly from unity to virtually zero.

Recent theoretical results suggest that it is possible to derive up to a good degree of approximation the full thermodynamical properties of planetary atmospheres from the coarse observations of radiative fluxes at the TOA and surface (Lucarini et al., 2011). In this work, we have pursued this line of research. This has been done by projecting the results obtained in exploring a 2D parametric space onto a single variable, thus constructing approximate parameterisations. We have found that the surface temperature and the emission temperature provide fairly good predictors of thermodynamic quantities representing the out-of-equilibrium

characterisation of the system. In some sense, such parameterisations provide an equation of state of the system.

The results discussed in this paper support the adoption of new diagnostic tools based on non-equilibrium thermodynamics for analysing the fundamental properties of planetary atmospheres. We do not claim that our results are in any sense universal. However, our results support the fact that such parameterisations are in principle possible. In order to modestly expand the scope of our paper, we have investigated how our findings change when considering a slower rotation rate. As is well known, doubling the length of the day has serious impacts on the global circulation of the atmosphere. Nonetheless, all of our approximate relations are only very weakly affected by such a radical change to the system. This is far from being a conclusive proof of universality, but does indeed show a degree of robustness for our results.

The reasons why these parameterisations work so well, and why this is especially the case for globally averaged surface temperature definitely require further investigations. These results reinforce the relevance of the globally averaged surface temperature as a meaningful thermodynamical quantity of the climate system, beyond its obvious practical importance. Moreover, they pave the way for the possibility of practically deducing fundamental properties of planets in the habitable zone from relatively simple observables. Future investigations will analyse more systematically how robust our findings are with respect to changes in relevant orbital parameters.

Acknowledgments

This work was supported by the EU-FP7 ERC Grant NAMASTE. The authors would like to thank E.T. Cartman, K. Fraedrich, P. Hau-schildt, F. Lunkeit and F. Ragone for their comments and insightful discussions.

References

- Arnol'd, 1992. *Catastrophe Theory*, third ed. Springer, Berlin.
- Bonfils, X. et al., 2013. The HARPS search for southern extra-solar planets *, **, *** XXXI. The m-dwarf sample. *Astron. Astrophys.* 549, A109, 1–75.
- Budyko, M.I., 1969. The effect of solar radiation variations on the climate of the Earth. *Tellus* 21, 611–619.
- Caballero, R., Langen, P.L., 2005. The dynamic range of poleward energy transport in an atmospheric general circulation model. *Geophys. Res. Lett.* 32, L02705. <http://dx.doi.org/10.1029/2004GL021581>.
- DeGroot, S., Mazur, P., 1984. *Non-Equilibrium Thermodynamics*. Dover.
- Donohoe, A., Battisti, D., 2011. Atmospheric and surface contributions to planetary albedo. *J. Clim.* 24, 4402–4418.
- Donohoe, A., Battisti, D., 2012. What determines meridional heat transport in climate models? *J. Clim.* 25, 3832–3850.
- Dvorak, R., 2008. *Extrasolar Planets*. Wiley-VHC.
- Eliassen, E., Mächenhauer, B., Rasmussen, E., 1970. On a Numerical Method for Integration of the Hydrodynamical Equations with a Spectral Representation of the Horizontal Fields. Report No. 2, Inst. of Theor. Met., University of Copenhagen.
- Fraedrich, K., 1979. Catastrophes and resilience of a zero-dimensional climate system with ice-albedo and greenhouse feedback. *Q. J. R. Meteorol. Soc.* 105, 147–167.
- Fraedrich, K., Lunkeit, F., 2008. Diagnosing the entropy budget of a climate model. *Tellus A* 60 (5), 921–931.
- Fraedrich, K., Jansen, H., Luksch, U., Kirk, E., Lunkeit, F., 2005. The planet simulator: Towards a user friendly model. *Meteorol. Z.* 14, 299–304.
- Gallavotti, G., 2006. *Encyclopedia of Mathematical Physics*. Elsevier, pp. 530–539 (chapter 'Nonequilibrium statistical mechanics (stationary): Overview').
- Ghil, M., 1976. Climate stability for a Sellers-type model. *J. Atmos. Sci.* 33, 3–20.
- Gough, D., 1981. Solar interior structure and luminosity variations. *Solar Phys.* 74, 21–34.
- Held, I.M., Soden, B.J., 2006. Robust responses of the hydrological cycle to global warming. *J. Clim.* 19, 5686–5699.
- Hoffman, P., Schrag, D.P., 2002. The Snowball Earth hypothesis: Testing the limits of global change. *Terra Nova* 14, 129–155.
- Hoffman, P.F., Kaufman, A.J., Halverson, G.P., Schrag, D.P., 1998. A Neoproterozoic Snowball Earth. *Science* 281, 1342–1346.
- Holton, J., 2004. *An Introduction to Dynamic Meteorology*. Elsevier.
- Johnson, D., 1997. "General coldness of climate" and the second law: Implications for modelling the Earth system. *J. Clim.* 10, 2826–2846.

- Johnson, D., 2000. Entropy, the Lorenz energy cycle, and climate. In: Randall, D.A. (Ed.), *General Circulation Model Development: Past, Present and Future*, International Geophysics Series, vol. 70. Academic Press, New York, pp. 659–720.
- Jones, C. et al., 2005. Systematic optimization and climate simulation of FAMOUS, a fast version of HadCM3. *Clim. Dynam.* 25, 189–204.
- Kasting, J., 2009. *How to Find a Habitable Planet*. Princeton University Press.
- Kennedy, M.J., Runnegar, B., Prave, A.R., Hoffman, K.H., Arthur, M.A., 1998. Two or four Neoproterozoic glaciations? *Geology* 26, 1059–1063.
- Kuo, H., 1965. On formation and intensification of tropical cyclones through latent heat release by cumulus convection. *J. Atmos. Sci.* 22, 40–63.
- Kuo, H., 1974. Further studies of the parametrisation of the influence of cumulus convection on large-scale flow. *J. Atmos. Sci.* 31, 1232–1240.
- Lacis, A., Hansen, K., 1974. A parameterisation for the absorption of solar radiation in the Earth's atmosphere. *J. Atmos. Sci.* 31, 118–133.
- Landau, L., Lifshitz, E., 1980. *Statistical Physics, Part 1*. Pergamon, Oxford.
- Laursen, L., Eliassen, E., 1989. On the effect of the damping mechanisms in an atmospheric general circulation model. *Tellus* 41A, 385–400.
- Li, J., Chylek, P., 2012. Atmospheric entropy. Part I: Climate dissipation structure. *J. Clim.* 25, 3173–3190.
- Liepert, G., Previdi, M., 2012. *Environ. Res. Lett.* 7, 014006. <http://dx.doi.org/10.1088/1748-9326/7/1/014006>.
- Lorenz, E., 1967. *The Nature and Theory of the General Circulation of the Atmosphere*, vol. 218.TP.115. World Meteorological Organization.
- Louis, J., 1979. A parametric model of vertical eddy fluxes in the atmosphere. *Bound. Layer Meteorol.* 17, 187–202.
- Louis, J., Tiedke, M., Geleyn, J., 1981. A short history of the PBL parametrization at ECMWF. In: *Proceedings of the ECMWF Workshop on Planetary Boundary Layer Parametrization*, 25–27 November, pp. 59–80.
- Lucarini, V., 2009. Thermodynamic efficiency and entropy production in the climate system. *Phys. Rev. E* 80, 021118, 1–5.
- Lucarini, V., Ragone, F., 2011. Energetics of climate models: Net energy balance and meridional enthalpy transport. *Rev. Geophys.* 49. <http://dx.doi.org/10.1029/2009RG000323>.
- Lucarini, V., Fraedrich, K., Lunkeit, F., 2010a. Thermodynamics of climate change: Generalized sensitivities. *Atmos. Chem. Phys.* 10, 9729–9737.
- Lucarini, V., Fraedrich, K., Lunkeit, F., 2010b. Thermodynamic analysis of Snowball Earth hysteresis experiment: Efficiency, entropy production and irreversibility. *Q. J. R. Meteorol. Soc.* 136, 1–11.
- Lucarini, V., Fraedrich, K., Ragone, F., 2011. New results on the thermodynamic properties of the climate. *J. Atmos. Sci.* 68, 2438–2458.
- Marotzke, J., Botzet, M., 2007. Present-day and ice-covered equilibrium states in a comprehensive climate model. *Geophys. Res. Lett.* 34, L16704. <http://dx.doi.org/10.1029/2006GL028880>.
- Orszag, S., 1970. Transform method for calculation of vector coupled sums. *J. Atmos. Sci.* 27, 890–895.
- Paoletti, S., Rispoli, F., Sciubba, E., 1989. Calculation of exergetic losses in compact heat exchanger passages. *ASME AES* 10 (2), 21–29.
- Pascale, S., Gregory, J., Ambaum, M., Tailleux, R., 2011a. Climate entropy budget of the HadCM3 atmosphere–ocean general circulation model and FAMOUS, its low-resolution version. *Clim. Dynam.* 36 (5–6), 1189–1206.
- Pascale, S., Gregory, J., Ambaum, M., Tailleux, R., 2011b. A parametric sensitivity study of entropy production and kinetic energy dissipation using the FAMOUS AOGCM. *Clim. Dynam.* <http://dx.doi.org/10.1007/s00382-011-0996-2>.
- Peixoto, J.P., Oort, A., 1992. *Physics of the Climate*. Springer-Verlag, New York.
- Peixoto, J., Oort, A., de Almeida, M., Tomé, A., 1991. Entropy budget of the atmosphere. *J. Geophys. Res.* 96, 10981–10988.
- Perryman, M., 2011. *The Exoplanets Handbook*. Cambridge University Press.
- Pierrehumbert, R., 2005. Climate dynamics of a hard Snowball Earth. *J. Geophys. Res.* 110, D01111. <http://dx.doi.org/10.1029/2004JD005162>.
- Pierrehumbert, R., Abbot, D., Voigt, A., Koll, D., 2011. Climate of Neoproterozoic. *Annu. Rev. Earth Planet Sci.* 39, 417–460.
- Pujol, T., 2003. Eddy heat diffusivity at maximum dissipation in a radiative–convective one-dimensional climate model. *J. Meteorol. Soc. Jpn.* 81 (2), 305–315.
- Read, P., 2011. Dynamic and circulation regimes of terrestrial planets. *Planet. Space Sci.* 59, 900–914.
- Seager, S., 2010. *Exoplanets*. University of Arizona.
- Saltzman, B., 2002. *Dynamic Paleoclimatology*. Academic Press, New York.
- Sasamori, T., 1968. The radiative cooling calculation for application to general circulation experiments. *J. Appl. Meteorol.* 7, 721–729.
- Sellers, W.D., 1969. A global climate model based on the energy balance of the Earth–atmosphere system. *J. Appl. Meteorol.* 8, 392–400.
- Slingo, A., Slingo, J., 1991. Response of the national center for atmospheric research community climate model to improvements in the representation of clouds. *J. Geophys. Res.* 96, 341–357.
- Stephens, G., 1978. Radiation profiles in extended water clouds. II: Parametrization schemes. *J. Atmos. Sci.* 35, 2123–2132.
- Stephens, G., Ackermann, S., Smith, E., 1982. A shortwave parametrization scheme. *J. Atmos. Sci.* 41, 687–690.
- Stone, P., 1978. Baroclinic adjustments. *J. Atmos. Sci.* 35, 561–571.
- Voigt, A., Marotzke, J., 2011. The transition from the present-day climate to a modern Snowball Earth. *Clim. Dynam.* 35, 887–905.
- Voigt, A., Abbot, D.S., Pierrehumbert, R.T., Marotzke, J., 2011. Initiation of a Marinoan Snowball in a state-of-the-art atmosphere ocean general circulation model. *Climate Past* 7, 249–263.